Chapter 5: Agent-Based Model

5.1 Model Conceptualization and Rationale

Building on Chapter 3, it becomes evident that the spatial and environmental distribution of Iron Age IIA sites favours the autochthonous interpretative model. As summarised at the end of Chapter 3, and demonstrated to some extent in the sample survey presented in Chapter 4, I suggest that the archaeological record of the Iron Age IIA in the Negev Highlands should be treated as a palimpsest of repeated short-term occupations by mobile pastoral groups throughout the timespan the system has been active, and that the pattern reflects both environmental and cultural considerations. This reconstruction will henceforth be termed the 'palimpsest model'.

The palimpsest model is difficult to test in traditional archaeological investigations. The seasonal mobility that drives palimpsest formation operates below the resolution of traditional dating methods, which resolve the record into blocks of decades at best, and more often centuries. Therefore, this model also cannot be tested directly in the field, given limitations in identifying both the number and exact duration of individual occurrences. However, it is possible to gain important insights regarding the observed archaeological record and the mechanisms that led to is formation through a series of computerised simulations that model locational decision making of participating social agents. As archaeologists, we constantly build models, as we routinely infer cultural and historical reconstructions from bodies of evidence such as ceramic distributions and frequencies (Lake, 2015 and references therein). A model simulation is a more formal, experimental version of this line of inference: "*a scientific technique in which a real-world system is represented in a simplified way and can be experimented on*" (Romanowska & Scherjon, 2023). In archaeology, such simulations are used for tactical method-testing, theory-building, and hypothesis-testing (Lake, 2014).

Agent-Based Modelling (ABM) is a simulation method well suited to modelling bottom-up dynamics that give rise to large-scale patterns, a phenomenon known as emergence (Lake, 2015; Romanowska et al., 2021). This bottom-up modelling is achieved through the actions of social units, or agents, which in this case were taken to represent households (see in detail below). ABM treats each agent as a decision-making entity that

follows a set of local rules (for example, where to camp or when to move). Simulation lets these household agents act and respond to their environment as well as to other social agents, both synchronically and diachronically, over many time steps (in this case, annual). The regional pattern of interest, the accumulated archaeological record (or palimpsest), is expected to emerge from the agents' behaviour if the model is successful.

Alternative approaches such as statistical models (e.g., regression or Maximal Entropy [MaxEnt]; compare e.g., Wachtel et al., 2018; Cortell-Nicolau et al., 2025 for evaluation of different methods) are fundamentally static, focused on end states rather than processes. By treating the archaeological record as fixed outcome of fixed predictors, temporal processes or feedback loops between human agents and environmental processes are difficult to identify (Romanowska, 2015; Verhagen & Whitley, 2020). Moreover, explicit computational models, and especially ABM, force researchers to specify their assumptions and test their logical consequences (Romanowska et al., 2019; Lake, 2020), thus providing a useful tool for hypothesis testing.

The current model aspires to answer the following question: **Can autochthonous nomadic processes, operating through temporal mobility and environmental feedbacks, sufficiently generate patterns similar to the observed Iron Age IIA archaeological record?** Accordingly, the simulation built here is of the hypothesis-testing type, though with a slight nuance. It sets out to test whether the dynamic processes and behaviours the palimpsest model envisions can generate a record resembling the one observed, thereby establishing it as a plausible scenario. It rests on a small set of assumptions and behavioural rules, detailed below.

- Nomadic groups, with varying degrees of mobility, characterised the Negev Highlands population during the Iron Age IIA and were responsible for the observed patterns (i.e., the regional archaeological record).
- At least part of the population moved seasonally between the Arabah and the Negev Highlands, as well as within the latter region.
- Subsistence economy, characterised as multi-resourced nomadism (compare Rosen, 2017), relied on primarily on pastoralism, manufacture and trade of copper and possibly other desert goods, and opportunistic agriculture.

- Stone-built structures of all types, including enclosed compounds, represent seasonal rather than permanent occupation.
- Tell el-Qudeirat (Kadesh Barnea) functioned as the central hub within the Negev Highlands system as the primary fresh-water oasis in the region (see Ben-Dor Evian, 2017 and references therein).
- The population is held constant throughout the simulation; while households may die, the populations stays stable by replacing those.
- The climate was similar to the present climate, as reflected in paleoclimatic studies (e.g., Goldsmith et al., 2023; Langgut & Finkelstein, 2023).
- This socio-economic system described above persisted for at least 75 years (the minimum span of Iron Age IIA using the 'low chronology') and, based on the radiocarbon evidence, probably considerably longer (see Section 1.4).

Based on these assumptions, the model aims to simulate the accumulation of nomadic activity and to test whether is translates into a palimpsest resembling the archaeological record. The decision-making units in the model (agents) are nomadic households, each of which is composed of an aggregation of nuclear families (see Section 5.2.4). Each household manages its location through a set of cultural and environmental considerations (detailed below, see Section 5.3.1), derived from the assumptions above. The annual repositioning of the household creates a cumulative pattern.

In order to systematically model the nomadic behaviour in the ABM, a set of 'behavioural rules' are operationalised:

- Agents select a location within the landscape once a year (representing either an inter-regional movement, e.g., annual cycle between the Negev Highlands and the Arabah Valley, or relocation within the former).
- Agents establish a loose territory round the selected location, that is taken to reflect kin-based territorial behaviour common among pastoral nomadic systems, and disperse the nuclear families within this territory (see details in Section 5.3.1).

- The agents' presence causes degradation of environmental resources around selected localities (through pasture, opportunistic agriculture), but these resources recover over time of abandonment of these localities.
- Agents learn from past experiences and form loose attachments to previously occupied locations and territories, especially since stone-built structures are built in these localities and can be easily re-used.
- Enclosed compounds are taken to express a form of household investment of accumulated surplus from economic activities, that function as communal kin-based, multi-purpose territorial centres. I hypothesize that these centres served purposes such as seasonal storage, social gatherings, and perhaps also exchange (see Chapter 6).

These operationalised rules are set this way not because they are thought to truly and fully represent how the population of the Negev Highlands Iron Age IIA behaved, but as general, simplified, tentative behaviours that are taken to be broadly inline with the assumptions detailed above.

One important principal that underlies several of these behavioural 'rules' is carrying capacity. In ecology, this concept is defined as the population that a given area can support (Hartvigsen, 2013). Carried into archaeology the concept has to be carefully used, because, unlike other animals, people can substantially alter a landscape's capacity, e.g., by digging cisterns, fertilising soils, or through herd management strategies (Glassow, 1978; see Caple & Løvschal, 2025). The model therefore uses the concept of carrying capacity in a generalised sense, as the environmental value it assigns to each location, composed of stable (e.g., topography) as well as annually changing (e.g., rainfall) factors, and the effect left by previous activity in the same locality (see Sections 5.3.1 and 5.3.2).

The model is, like all models, a simplified representation of the real-world system (Romanowska et al., 2019). For instance, it does not represent demographics or internal tensions of the people within each unit, although these surely influenced where groups camped or how clustered they chose to be. Environmental processes, such as resource degradation and recovery, are captured by deliberately simple functions rather than full

ecological models, which lie beyond the scope of this work. Because both household decisions and year-to-year conditions incorporate an element of chance (stochasticity), no two runs of the model produce identical results. Therefore, the model's behaviour is characterised from many repeated runs. Uneven preservation and survey coverage also bear on how closely the simulated pattern can be expected to match the real record.

This set of assumptions and rules is what leads the model to test the accumulation of nomadic activity into a palimpsest; the sections that follow specify each component before turning to the model's calibration and results.

5.2 Model Structure and Components

This section sets out the model's architecture: the landscape it runs on, from the grid of cells to the environmental layers households read; the household agents themselves; and the timescale over which they act. The model is implemented in Python with the Mesa agent-based modelling framework (ter Hoeven et al., 2025) and available in this GITHUB REPOSITORY.

5.2.1 Spatial and Environmental Representation of the Negev Highlands

5.2.1.1 Spatial Specifications

The model represents the region as a grid of square cells. The idea is to represent the continuous environmental and anthropogenic data that households make their decisions upon as raster layers. Each cell is 250 × 250 m (62.5 dunam), across a grid of 318×280 cells. This size was set to balance computational efficiency and three spatial limitations: the positional error of survey coordinates (roughly 100–500 m in most surveys), the spread of Iron Age IIA sites (whose remains can extend over some 400 m; Rothenberg, 1967), and the resolution of the available environmental datasets.

The modelled area follows the natural topographic limits of the Negev Highlands (see Section 1.3) where it can. The exception is the western edge that is cut at the modern Israel–Egypt border due to data limitations (see Section 3.2 above), as the model can only represent terrain for which comparable environmental data exist. That border therefore cuts the study area short of the natural western extent of the Negev Highlands, leaving 48 Iron Age IIA sites (9.4% of the dataset) outside the spatial comparison (Figure 5.1).

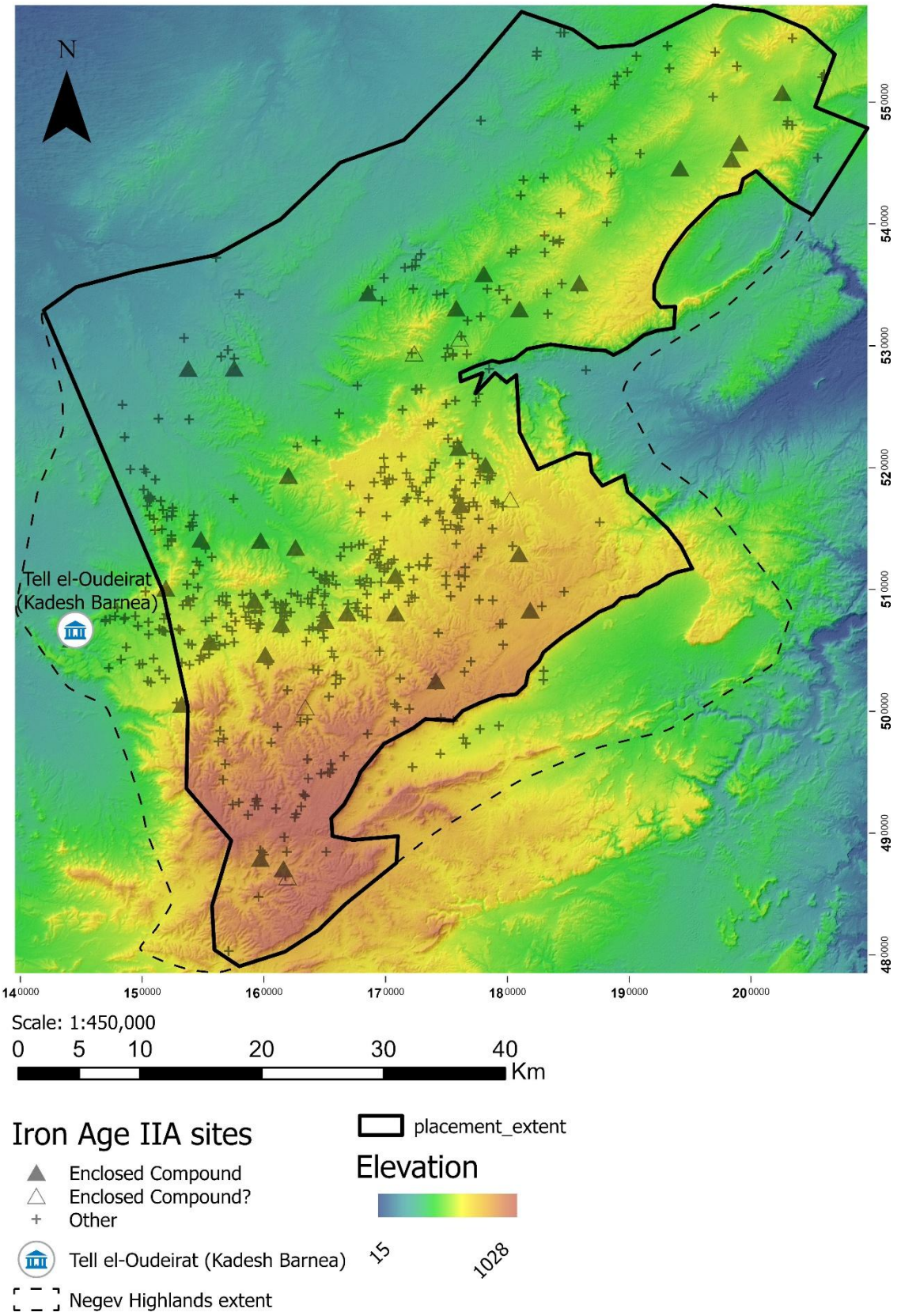


Figure 5.1. Study area and model spatial extent. Map showing the geographic boundary of the Negev Highlands (dashed line) and the placement extent of the model (solid line). Iron Age IIA sites are symbol-coded by type: enclosed compounds (filled triangles), uncertain enclosed compounds (hollow triangles), and other architectural sites (crosses). Hillshade basemap derived from Jaxa Alos Palsar DEM; elevation in metres (ITM).

Two choices about the grid's edges follow from the same concern with realism. Space is bounded, not "toroidal": unlike many ABMs, where an agent leaving one edge reappears on the opposite side (as in Pac-Man), households here cannot (Lake, 2020). Since cells at the very edge would otherwise lack neighbours, a buffer zone extends beyond the valid placement area — environmental processes such as degradation and resource availability are computed across this wider zone, while a binary "place raster" confines actual camps to the Israeli part of the Negev Highlands. Both choices keep the boundary from affecting the model's outcome.

All layers share the Israel Transverse Mercator (ITM) projection, consistent with the database (Appendix 1) and preserving the same distances for movement and territory calculations as in chapter 3. The environmental layers themselves divide into permanent and annually varying components, described below.

5.2.1.2 Environmental and Anthropogenic Variables

The model represents the environment through layers classified along two axes: ecological versus anthropogenic, and permanent versus annual (Table 5.1). This separation lets the model hold the region's stable base environment constant while capturing the year-to-year variability characteristic of arid environments, together with the households' own impact on the landscape. Ultimately all of these layers are combined into a single weighted suitability score for each cell, which is what households read when choosing where to settle; the paragraphs below describe each layer in turn before setting out how they are weighted and summed.

Before the layers can be combined into that weighted sum, they must be placed on a comparable footing, since they arrive in incompatible units — millimetres of rain, degrees of slope, hours of travel — and cannot simply be added. Each is therefore rescaled onto a common 0–10 "suitability" scale, on which 0 marks conditions unusable for settlement and 10 the most favourable. The rescaling uses a fuzzy-membership function (Fusco & de Runz, 2020): rather than sorting each cell into discrete classes, it assigns a continuous graded score that rises or falls smoothly with the underlying value. In doing so, the function captures how agents are assumed to respond to the resource. For instance, a zone with better rainfall in a given year is assumed to be more attractive, but only up to a point: above a certain threshold, additional rainfall no longer adds to its appeal. The

shape of each function — where it begins to rise, how steeply, where it levels off — was set to match how that factor bears on settlement suitability; full specifications are given in Appendix 5.1.

Permanent Environmental Variables

Five permanent environmental variables remain fixed for the whole run and represent the region's stable backdrop of geography and long-term climate. Mean annual rainfall is interpolated across the region from Israeli Meteorological Service (IMS) station records by inverse distance weighting (IDW). IDW estimates the value at each cell as the average of the surrounding stations, weighted according to distance, so that nearer stations count for more. Slope is derived from the JAXA ALOS PALSAR digital elevation model (see Section 3.2). Proximity to arable soils is based on Pleistocene loess and Hatzeva Formation sandy soils, mapped from Israeli Geological Survey data (Section 3.2), and distance to permanent water on springs and wells compiled from topographic and other sources (Section 3.2). The fifth layer, vegetation type, sets each cell's pastoral value (based on principles & maps from Danin, 1983; Ron et al., 2024).

Explicit vegetation layer enters the model although Chapter 3 used soils alone as a proxy for pasture and opportunistic agriculture. The reason is that the model needs to quantify pasture productivity across the whole landscape as households graze throughout the landscape. Vegetation type supplies a grazing value for every cell (Section 5.3.2); the loess soils themselves register as some of the more productive pasture within it.

Distance in the model is measured in walking time. Tobler's hiking function converts slope into walking speed, giving steeper slopes more time to cross (Tobler, 1993; I. Herzog, 2020; see Section 3.3). The same treatment applies to every distance-based layer — distance to water here, and the annual arable and central-site layers introduced below.

Annual Environmental Variables

Rainfall in the Negev Highlands swings widely from year to year, roughly 30 to 160 mm (Armon et al., 2019; see Section 1.3), and this variability, more than the long-term average, is what a mobile population must cope with. The model therefore treats annual

rainfall as stochastic - each simulated year draws its rainfall at random from a set of observed years. This choice responds to two facts: the wide variability just noted, and the absence of any direct evidence for how much rain fell in specific years during the Iron Age IIA. Drawing independently each year from measured present-day rainfall reproduces a realistic spread of good and bad years without pretending to reconstruct particular ones. In practice, each year's rainfall is taken from a randomly chosen observed year (August–July) in the IMS record, restricted to years with at least 20 active stations so that spatial coverage is reliable, then interpolated (IDW) and rescaled on the same 0–10 fuzzy scale as permanent rainfall.

Annual rainfall matters less on its own than through the two seasonal resources it governs, both exploited opportunistically by mobile groups: grazing and short-lived cultivation. Available pasture is the product of vegetation type and that year's rainfall. Cultivation is handled separately, and rests on how rainfall affects crops. Opportunistic agriculture depended on runoff draining into it from the slopes above, such runoff farming long documented in the region in later periods (Bruins, 2012; Garty et al., 2025, and references therein). Accordingly, for each loess or Hatzeva Formation patch the model reads the year's rainfall across the whole basin draining into it. If the rainfall is adequate on average and reliable enough across the basin the patch is marked seasonally cultivable. These cut-offs remain model-design choices (Appendix Table A5.1), but they are set around the annual-rainfall level below which cultivation models treat a year as a drought, and the crop as failing (on the order of 65 mm; Garty et al., 2025). Cultivable patches are then measured by cost-distance, like the other distance layers.

Permanent Anthropogenic Variable

The one permanent anthropogenic variable in the model is proximity to the Tell el-Qudeirat–Faynan corridor. According to the palimpsest model, the activity in the Negev Highlands belonged to a multi-resource pastoral economy in which moving people for labour into and copper out of the Arabah was a central occupation (Section 5.1). Therefore, households would have found locations near the broad east–west axis linking the Faynan copper district to Tell el-Qudeirat attractive to a certain extent as shown in Chapter 3 (Section 3.4.3). In practice it is computed as the walking-time cost-distance to the corridor (Figure 3.11; only the red parts), using the same procedure applied to the

permanent water sources, then rescaled onto the shared 0–10 suitability scale (transformation parameters in Appendix 5.1).

Dynamic Anthropogenic Variables

The model's two dynamic anthropogenic variables represent the accumulated imprint of past human activity: resource exhaustion and the reuse of built sites.

Resource exhaustion is a key process in the model, as occupying a site (a cell in the model) is assumed to have a negative effect on its immediate environment. For example, grazing puts pressure on the local vegetation, whose regrowth is slow to begin with in such an arid environment. This pressure is also uneven: the negative effect is assumed to decline with distance from the camp. The model carries this forward as a running sum of degradation: each year's impact is added to earlier ones, with recent occupation weighing more than older occupation. The sum is then smoothed across the neighbouring cells (focal statistics), producing a zone of impact around each occupied location (Section 5.3.1).

By contrast, occupying a site is also assumed to add to the built infrastructure there, raising its value for future use. Households are therefore drawn back to locations already used, and above all to standing enclosed compounds, which represent a more substantial investment. This pull grows with the intensity of past occupation and fades as a location sits unused. It reflects the tendency for familiar, already-developed locations to be reoccupied across generations.

Together these variables represent the dynamic character of this delicate environment. At any given time, some locations are overexploited while others have already recovered. This shifting patchwork of depletion and inherited investment is one of the temporal mechanisms at the core of the palimpsest reading.

Suitability Calculation

The model collapses the variables above into a composite score for each cell, which forms the basis for household site selection. Each variable is first rescaled onto the shared 0–10 scale, then combined into one overall suitability value through a weighted sum:

$$Suitability = \sum(W_i \times Parameter_i)$$

The weights (W_i) are the model's parameters, setting how strongly each variable pulls on the decision; their values are explored by calibration (Section 5.5).

Seven variables enter the sum: cost-distance to the Tell el-Qudeirat–Faynan corridor, cost-distance to permanent water, the region's long-term mean rainfall, slope, resource exhaustion, reuse potential, and the current year's rainfall. Available pasture and cultivable land are deliberately kept out. Both are derived from that year's rainfall (Annual Environmental Variables, above), so weighting them next to rainfall would overrepresent it (Table 5.1).

5.2.2 Household Agents

The model's main actor is the household, with its nuclear families as sub-agents. Working at this level is not a claim about how Negev Highlands society was actually organised (the record does not let us reconstruct that) but follows from what the model sets out to explain. The palimpsest builds up from a few decisions, made repeatedly: where to camp, whether to build, when to sell animals. Whatever the wider political and social arrangements of Negev Highlands inhabitants were, they matter less at this local level. Ethnographic studies agree that decisions of this local kind typically rest with the single family or household levels (Khazanov, 1984; Salzman, 2004; Hammer, 2025). The household is therefore the level at which the choices the model cares about were taken, which makes it the natural unit here, regardless of the exact social structure. "Household" is accordingly a modelling simplification: a tentative group of about ten nuclear families, not a reconstruction of any specific tribal organisation. These households are taken to have practised multi-resource nomadism, as argued in Chapter 3: pastoralism was one important strand alongside opportunistic agriculture and the production and movement of Arabah copper and other desert goods.

Each household needs a herd large enough to support it, and the model sizes that herd from what a person requires. The pastoral baseline (introduced in Section 5.2.4 below) is about 37 goats per person under a purely pastoral economy (Dahl & Hjort, 1976; Rosen & Finkelstein, 1992). Because the households here practised multi-resource nomadism rather than pure pastoralism, as noted above, the model lowers this to 18 goats per

person, about 50% of the pastoral-only figure. This is based on the assumption that a group leaning partly on trade and farming needed fewer animals per head to subsist. Rosen and Finkelstein point the same way, finding that the Sde Boqer area supported more people than herding alone could feed, so that its population must have drawn on external resources such as trade and copper. For a household of about 50-60 people this gives a herd of roughly 1,050 goats, the value around which each starting herd is drawn.

Each household has nine state variables (Table A5.5; how they are being used is detailed in 5.3):

- Livestock: the herd, counted in goats standing for all animals; the household's main productive asset, varying with time and environment (Section 5.3.2).
- Manpower: the working-age people available for all tasks, herding, movement, and building, initialised as whichever is larger, 35 people or one per 20 animals (as set in Section 5.2.4); it caps how much a household can do and build.
- Surplus: accumulated capital, built from herding and, implicitly, opportunistic farming and copper trade; drawn down in lean years and, when it runs high, may be spent on enclosed compound building (Sections 5.3.2-5.3.4).
- Camp location: the cell the household occupies in the current year, cleared at year's end and re-selected at the next.
- Territory: the block of cells around the camp that it grazes and ranges over, sized to local conditions and reset each year.
- Camp memory: a personal record of past camps, each kept with its location, the quality its close environs offered, and the year of the stay, and thinned over time.
- Enclosure memory: a longer, shared record of enclosed compounds, both the household's own and others' it has come across.
- Personalised suitability: the household's own conception of the landscape. Its camp memory and past territories translated into location-specific bonuses and penalties laid over the shared environmental suitability.

- Crisis flag: whether the household has fallen into survival crisis, which triggers emergency livestock sales or acquisition.

The three that matter most, livestock, manpower, and surplus, carry the household's economy and shift year to year as it manages its herd (Section 5.3.2). The two memory variables let otherwise-identical households diverge in behaviour. How that remembered landscape enters the yearly location decision is set out in Section 5.3.1.

Together, these variables drive four kinds of behaviour, each treated in Section 5.3: annual positioning (5.3.1), resource management (5.3.2), prosperity-based enclosure construction (5.3.3), and crisis response (Section 5.3.4) (Figure 5.2). Each behaviour follows from a household weighing its current state against the conditions around it.

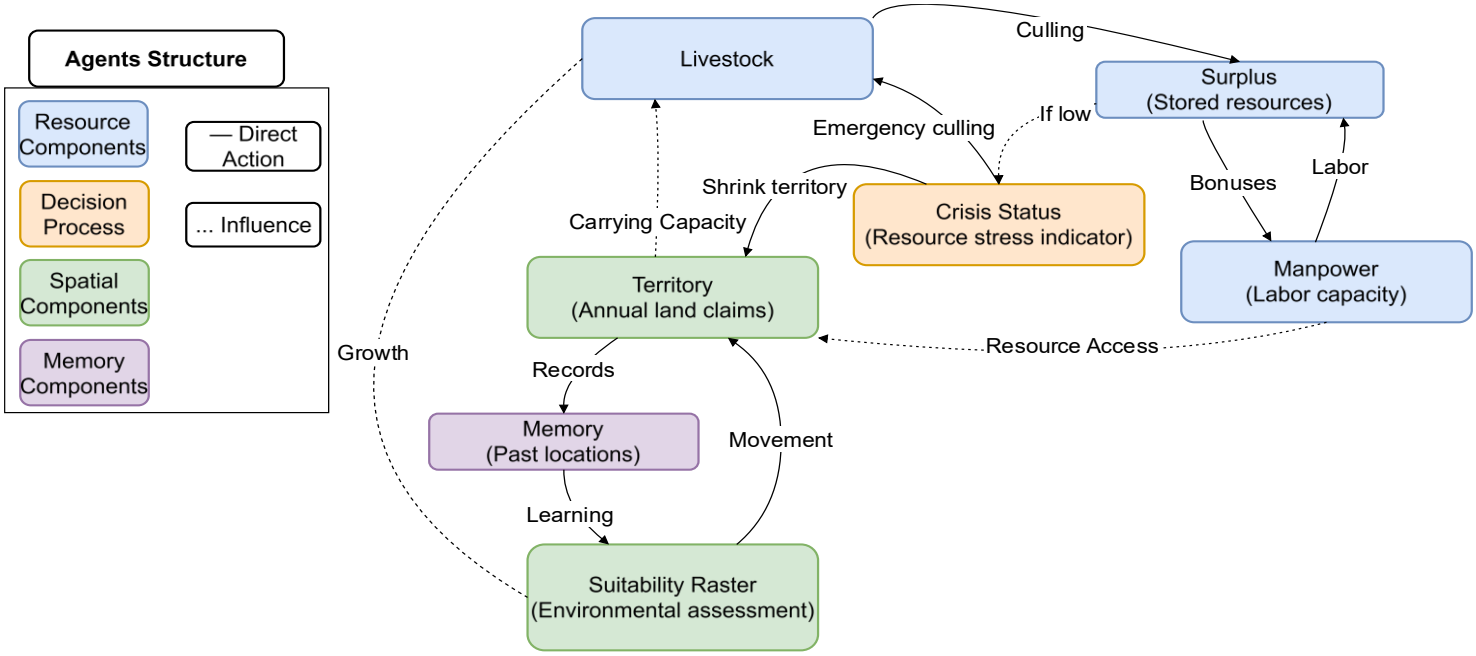


Figure 5.2. Household agent state variables and their interactions. Nodes are colour-coded by category: Resource Components (blue: Livestock, Surplus, Manpower), Decision Process (orange: Crisis Status), Spatial Components (green: Territory, Suitability Raster), and Memory Components (purple: Memory). Solid arrows indicate direct actions; dashed arrows indicate indirect influences.

5.2.3 Time Frame and Temporal Resolution

The reason behind the model's chosen time frame — 75 years, advancing one year at a time — is two-fold. First, the overall span adopts the minimum duration attributed to the phenomenon under the low chronology (Boaretto et al., 2010; Finkelstein & Piasetzky, 2011; see Section 1.4.1). Although the radiocarbon record suggests the activity in fact persisted considerably longer (Bruins & van der Plicht, 2007; Mamalya & Erickson-Gini, 2019), adopting the minimal length is a deliberate choice: if the palimpsest can plausibly emerge within the shortest span the record allows, a longer duration would only reinforce it. Second, the model's time steps are annual, a resolution fine enough to simulate the decisions and actions hypothesized to generate the archaeological record without being finer than necessary. Following the KISS principle (Keep It Simple, Stupid; Romanowska et al., 2021, and references therein), greater detail would add both uncertainty and computational cost.

Each annual step stands for a single episode of occupation within a given year in the Negev Highlands. The idea is that a household's location in the region is not fixed: from one step to the next it may move to a new camp, and the model does not specify where that move begins. The origin could be the Arabah, the household's own previous encampment within the region, or another area altogether. Representing relocation as a single move once per year is a simplification, since the real rhythm of movement is unknown, and probably would have been diverse. Hence, each step simulates the selection of a camp, based on that year's environmental conditions, each household's accumulated experience, and its coordination with the placement of neighbouring ones (Section 5.3.1). This repeated, step-by-step re-selection is key to generating the model's spatial dynamics.

Within each simulated year the model carries out a fixed sequence of steps, each of which sets the conditions for the next. It first updates the environmental variables for that year, then lets each household select a cell it finds most suitable and has it establish a territory and disperse its nuclear families within it. Then, it judges whether the resulting surpluses are enough to build or reuse an enclosed compound (Section 5.3.3), and finally it adjusts each household's livestock and demographics and accumulates the degradation that occupation leaves on the landscape. Running the steps in this order

produces a chain of effects: environmental conditions shape where a household settles; its location determines the resources it acquires; those resources drive its decision to build or reuse enclosed compounds (Figure 5.3); and the resulting occupation degrades those cells, lowering their suitability in later years.

Repeated over a full run, these yearly steps are what turn individual agents' decisions into an accumulated palimpsest, the pattern the model sets out to reproduce (Section 5.1). Favourable cells, revisited across many non-consecutive years, build up dense concentrations, while unattractive cells, used only occasionally, leave sparse traces. In this way, the final distribution superimposes many non-contemporaneous occupations.

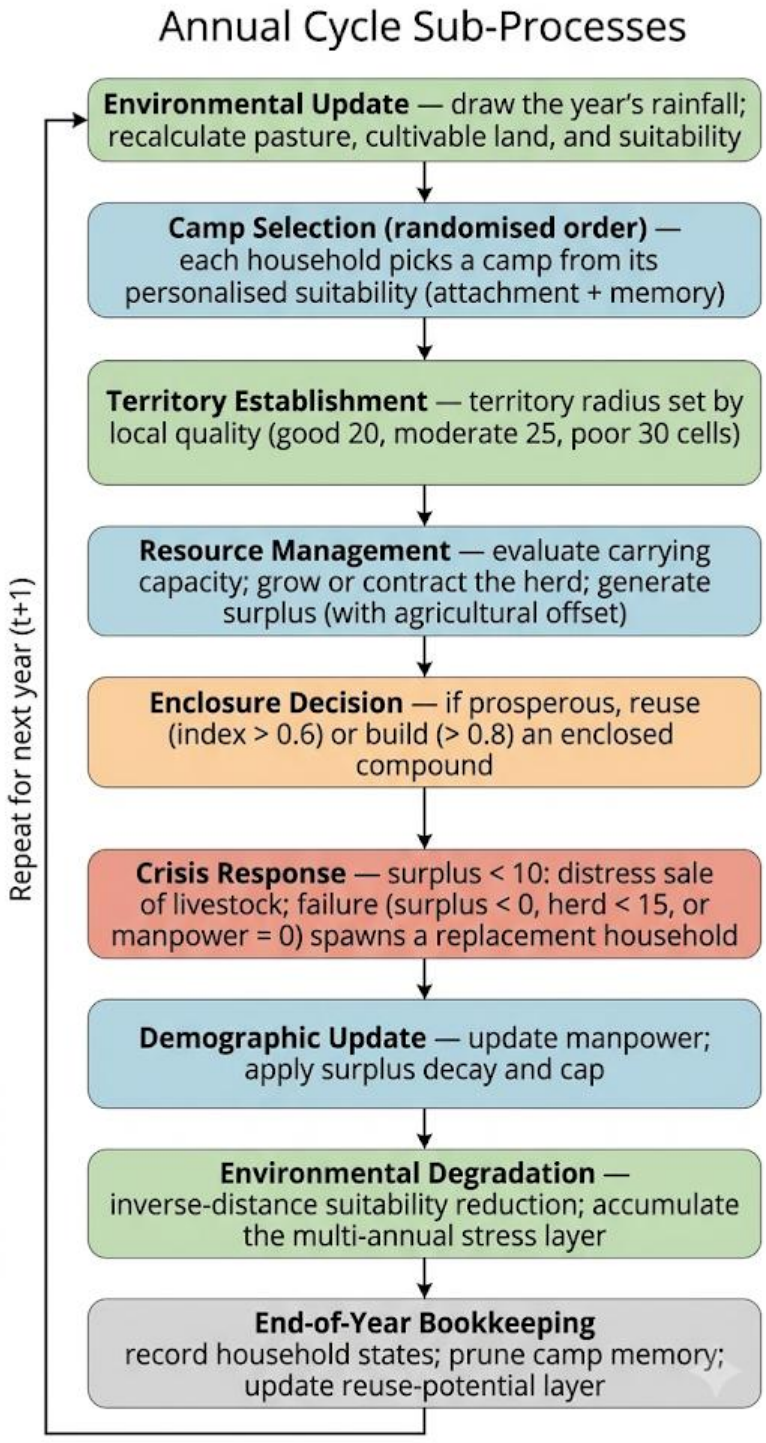


Figure 5.3. Annual cycle sub-processes. Eight sequential stages execute each simulated year: environmental update; household repositioning in randomised order; territory establishment; resource calculation; prosperity assessment and enclosure decision; crisis check; livestock and demographic adjustment; and end-of-year bookkeeping. Cycle repeats for t+1. See Figure 5.6 for outer loop context.

5.2.4 Model Initiation

Finally, a word on where a run begins. The idea is to simulate a population well below the region's carrying capacity, reflecting a deliberately conservative scale of activity. The simulation uses nine households, held constant across the run.

That figure comes from the grazing capacity of the Negev Highlands. Seligman et al.'s (1962) rangeland survey found the region supports on the order of 5–10 head of sheep per km^2 each year, the lower end in its more marginal reaches. Rosen and Finkelstein (1992) built on that survey to express the same capacity in human terms: converting small-stock to goats (about 1.7 goats per sheep) and adopting Dahl and Hjort's (1976) estimate that a six-person herding family needs some 223 goats a year to meet its caloric needs, roughly 37 per person, under a purely pastoral economy. On the marginal capacity of about 5.5 head per km^2, this comes to roughly 4 km^2 of ground per person, or on the order of 18 km^2 for a nuclear family of five to six people. The model takes this conservative, marginal figure as its baseline.

From there the number of households follow. A household in this model groups about ten such families and so needs on the order of 180 km^2. The study area of some 3,251 km^2 could therefore sustain roughly 18 households at carrying capacity. The model begins with half of these, nine households, again to be on the conservative side. The point is not to estimate an approximate population of the Iron Age IIA, but to take a conservative figure: if the palimpsest can form even from such sparse population, a denser one would only reinforce it. These nine households begin with no accumulated experience or surplus and, in their first year (Year 0), choose their location on environmental suitability alone (Section 5.3.1); thereafter replacements take the place of any household that fails (Section 5.3.4). Full parameter values are listed in Appendix 5 (Table A5.4).

5.3 Agent Decision-Making and Behaviours

This section details the processes taking place in the simulation, either by agents' actions or as environmental ones. In principle, household's activity reduces to a handful of recurring decisions, each made afresh at each step in response to its environment and resources. The following processes are detailed below: (i) camp selection and its consequences, (ii) use of manpower and herds as resources and conversion to surplus,

(iii) the decision-making process leading to enclosed compounds building or reuse, and (iv) crisis management. The accumulation of many such local decisions, repeated across the run and across households produces the regional palimpsest the model sets out to explain (Section 5.1).

5.3.1 Camp Selection and Environmental Feedback

The annual decision of where to camp combines the environmental suitability of the landscape, which is the same for every household, with each household's own reading of it. In its first year (Year 0) a household has no past experience and chooses on environmental suitability alone (for suitability calculation see Section 5.2.1.2); from Year 1 onward it keeps that suitability as the starting point and adjusts it through its own experience: an attachment to territory and a memory of past camps. The choice itself is probabilistic: more suitable cells are likelier to be picked, but not certain to be.

The first form of experience, attachment to territory, represents a household's tendency to keep to a familiar part of the region rather than range across it solely on environmental basis (Section 5.2.2). This kind of spatial inertia is widely reported in pastoral systems, where a group's camps recur on the same ground, or close to it, from one year to the next (Avni, 1996; Galilee, 2013; Hammer, 2014; Meraiot et al., 2021). The model represents it as two bonuses added to the environmental suitability of each cell (Section 5.2.1.2), acting at two spatial scales.

The broader bonus draws a household toward the general area of its past activity. The model tracks a home-range centre, *C*, updated each year as a running average of camp positions:

$$C = 0.9\, C_{\text{prev}} + 0.1\, P$$

where *P* is the current year's territorial centre and C_{prev} the previous centre (in the first year, *C* is the first camp). The weighting shifts the centre only a tenth of the way toward each new camp, so it moves slowly and holds successive territories in the same general area. Each cell then receives a bonus:

$$B = 5.0 \cdot exp(-D^2 / (2 \cdot 50^2))$$

where *D* is the distance in cells from the cell to *C*, 5.0 is the maximum bonus on the 0–10 suitability scale, and 50 cells (12.5 km) sets its spread; the bonus is greatest at the centre and fades to near zero about 35 km away.

The finer bonus adds a sharper pull toward the ground used the previous year. Cells within 20 cells (5 km) of the previous year's territory receive a bonus of: $\frac{2.5}{(d+1)}$, where *d* is the distance in cells to the nearest territory cell, so the bonus is strongest at the territory itself and decays with distance from it.

Alongside this pull toward familiar territory, households hold memory of individual camps that affects their next choice of location. They prefer to return to camps which had good quality at the year of occupation and avoid bad ones. Each household therefore carries a record of the camps it and its nuclear families have occupied. Every entry holds three things: the camp's location, the environmental quality *q* it offered (on the same 0–10 scale as suitability), and the year of the stay.

That record adjusts the suitability of the remembered cells in three ways. A cell occupied in the current year is simply penalised, −0.5, making it less probable to be re-selected. For camps used in earlier years the adjustment takes the value of the remembered quality and fades with time, its weight being:

$$w = \frac{1}{\sqrt{y_\Delta}}$$

where y_Δ is the number of years since the stay, so a camp used two years ago has more weight than the same camp recalled from ten years back. A good camp of quality $q \geq 5$, adds a bonus that draws the household back:

$$A = q \times w \times 0.5$$

while a poor camp of $0 < q < 5$ subtracts a penalty:

$$A = -\left(\frac{1}{q}\right) \times w \times 0.5$$

as the remembered quality falls the penalty is higher, so the least rewarding camps are avoided most strongly.

Memory is imperfect and does not last indefinitely. At the start of each year a household prunes its record: camps younger than 15 years are kept, while older ones are dropped with probabilistic logic, the older they are, the chance to be forgotten increases. Therefore, few memories persist much beyond two decades. A household thus evaluates the environment with a limited, recency-weighted memory of past camps. Together with the territorial attachment above, these adjustments turn the shared environmental suitability into the personalised raster which each household uses to make his camp location decision.

These experiential adjustments give each household its own suitability raster, from which the year's camp is selected. The draw is probabilistic and weighted by suitability raised to the power of three, so a better cell is much more likely to be chosen while a poorer one keeps a slim chance. This transformation of the raster's values sharpens the preference without making it deterministic. The surface a household reads already carries the effect both of other agents occupying the area and of earlier occupation (Section 5.2.1.2, and detailed below), so currently and recently used ground has lower suitability, making it less likely to be selected as a camp.

Once a cell is chosen the household lays out a territory around it, reflecting the zone its herds will range over for the time of occupation. The territory is defined by a radius that widens as conditions worsen, from 20 cells (5 km) on favourable ground to 25 and 30 cells where local environmental quality is sub-moderate or poor. Put plainly, a household ranges more widely when resources are thin. Its nuclear families then disperse across this territory, each placed by the same logic, so the household's presence is spread over its territory rather than fixed at a single point. Each of these family camps, like the household's location, degradations down the ground around it, contributing to the environmental degradation detailed below.

Occupation leaves a lasting imprint on the environment in two opposing ways. It degrades the area a household uses, but it also makes that specific cell more likely to be chosen again as it has been invested in (e.g., by building stone structures). The first, degradation, is the model's central environmental process and the source of much of its temporal dynamics. Grazing, trampling, and the gathering of wood and water deplete a cell's resources. Overgrazing in particular, can exhaust the local vegetation of such arid

environment (compare Boles et al., 2019; Meroz et al., 2023 for Negev-specific study), which then recovers only slowly. Accordingly, cells occupied recently, whether by a household itself or by others, lose some of their suitability value. The model represents this degradation directly in the suitability surface: as a household establishes its camp, the cells it occupies and their neighbours lose suitability. This lowering of the overall environmental value has a direct effect on the next households seeking a place to set camp.

Degradation enters at two instances. The first is immediate. As soon as a household establishes its camp, the model lowers the suitability of the surrounding cells (Figure 5.4). The reduction is largest at the occupied cell and decays with distance from it, extending across the household's territory. The nuclear family camps within that territory impose the same reduction at a smaller scale, over a radius of about 11 cells (≈2.75 km) each. Because these reductions are written to the shared suitability surface the moment they occur, households positioned later in the same year face an already-degraded landscape and are less likely to make camp nearby. This is the coordination noted above, modelled implicitly, arising through environmental limitation.

The second instance is cumulative and spans years. Alongside the immediate reduction, the model maintains a stress layer (Section 5.2.1.2) that records the imprint of past occupation. Each year's occupation is added to this record, but recent occupation weighs more than earlier one:

$$S_t = 0.5 \times S_{t-1} + O_t$$

where S_t is the accumulated stress and O_t the current year's occupation. The influence of this stress fades over several years, functioning as the model's stand-in for the slow environmental recovery. The stress raster is then smoothed, each cell taking a distance-weighted average of its neighbours within about 20 cells (5 km), spreading stress into a zone around each once occupied cell. Finally, the smoothed stress is rescaled onto the 0–10 scale by a fuzzy function, so that cells far from any occupation score 10 and heavily occupied ones approach 0. This rescaled stress raster then enters the weighted suitability sum as the resource-exhaustion variable (Section 5.2.1.2).

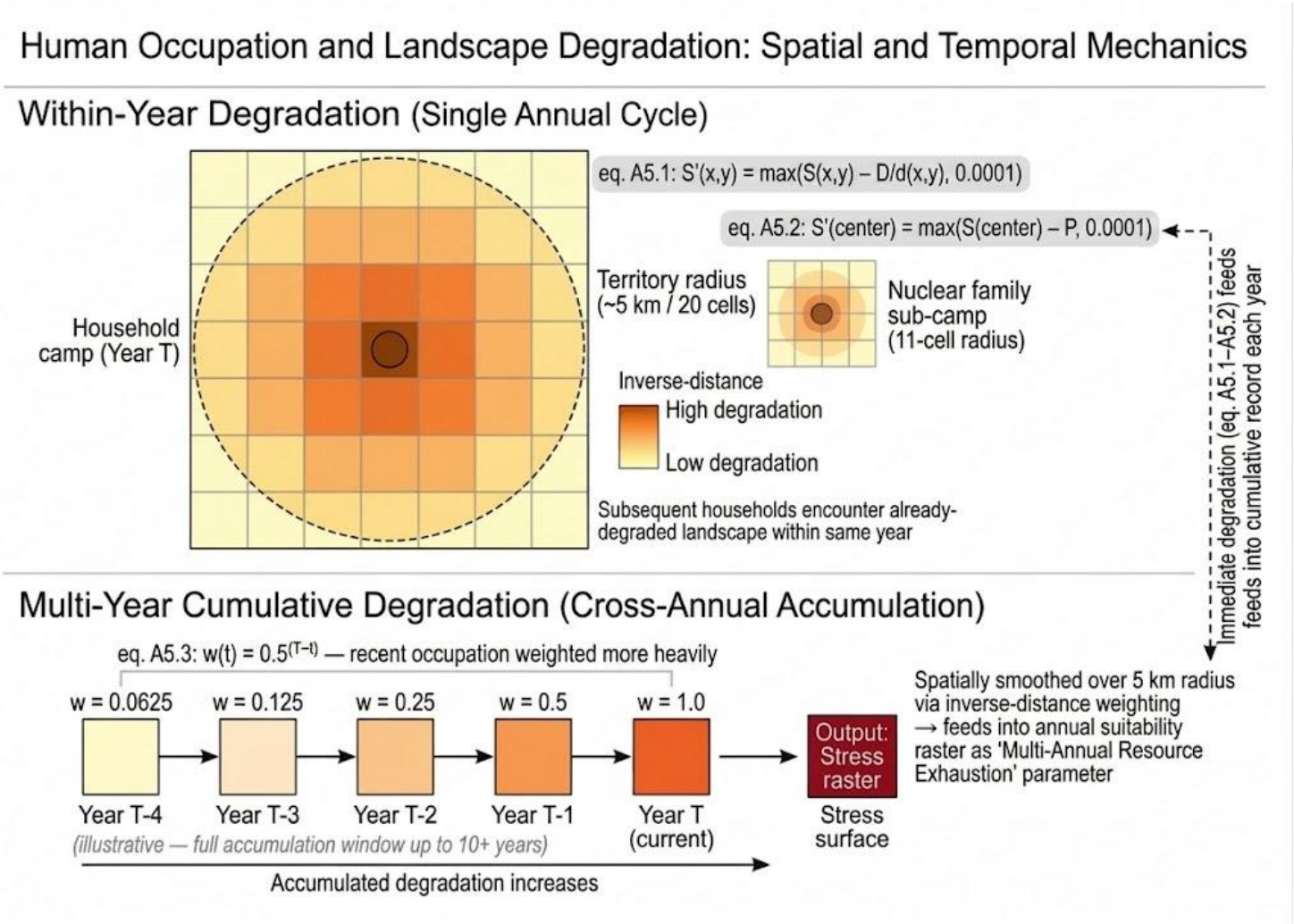


Figure 5.4. Spatial and temporal mechanics of landscape degradation. Top: within-year inverse-distance suitability reduction across household territory (~5 km radius; eq. A5.1) and nuclear family sub-camp (11-cell radius; eq. A5.2). Bottom: cross-annual accumulation weighted by temporal decay w(t) = 0.5^(T–t), spatially smoothed over 5 km and output as the Multi-Annual Resource Exhaustion parameter in the annual suitability raster (Table A5.1).

As noted above, the same record of past occupation also acts in the opposite direction. The reason is that each visit to a cell is assumed to add to the built infrastructure there, raising its value. Accordingly, alongside the exhaustion just described, the accumulated occupation of each cell enters the suitability surface a second time, now as a positive term that attracts households back to already-used locations (Section 5.2.1.2). Each occupied cell is rewarded in proportion to how intensely it was occupied (a household has a stronger effect than a nuclear family). Hence a site visited repeatedly, by households or nuclear families, gains a higher suitability value. An enclosed compound leaves a far heavier mark than an ordinary camp. It therefore exerts a strong and lasting pull, developed further in Section 5.3.3.

5.3.2 Resource Management and Surplus Calculation

A household's economy turns on three linked stocks: its herd, its manpower, and its surplus. Surplus is stored wealth, the produce a household accumulates beyond what it needs to feed itself in a given year, held as an abstract quantity rather than in any one commodity. It is what a household later draws on to build and maintain enclosed compounds (Section 5.3.3) and to weather poor years (Section 5.3.4). Each simulated year the household manages its resources in two steps. First, the herd grows or contracts according to the surrounding pasture. Second, the household converts the livestock beyond its subsistence needs into surplus, from which the year's consumption is then deducted.

The livestock variable is a dynamic one. First, the model evaluates the pasture across the territory and converts it into a carrying capacity, the number of livestock it can sustain:

$$Capacity = n \times 1.125 \times \left(\frac{PV}{5}\right)$$

where *n* is the number of cells in the territory, *PV* their mean pasture value (affected by previous occupations, Section 5.3.1) on the 0–10 scale, and 1.125 reflects goats-per-cell ratio that medium pasture (*PV* = 5) supports, keeping with the rangeland capacities set out in Section 5.2.4. Comparing this capacity with the current herd sets the year's trajectory. Where the environment offers comfortably more than the herd needs (capacity above about 1.1 times the herd), the herd grows; when it falls short (capacity below about 0.9 times the herd), the herd contracts. Between these bounds the herd holds roughly

steady. Two edge cases temper these dynamics - small herds recover faster (below 500), as they are easier to manage; very large herds (above 1,000) present the household with diminishing marginal returns, reflecting their unwieldiness.

Beyond subsistence, the herd is the source of surplus: a household sets aside the livestock it needs to sustain itself and draws surplus from the rest. That subsistence need is calculated as:

$$Need = M \times 18 \times (1 - a)$$

where *M* is the household's manpower and 18 the animals per person required for subsistence (Section 5.2.2). *a* represents a bonus from opportunistic cultivation. Where well-watered arable patches are present in the household's territory (its extent set by that year's rainfall in the basin above the patch), it substitutes for part of the herd, reducing the livestock need by up to 40%, so households on better farmland need fewer animals to get by (compare Rosen & Finkelstein, 1992; Günther et al., 2021 who modelled similar systems).

The herd yields surplus in two ways, and how much turns first on how many animals a household keeps in hand. Beyond the bare subsistence *Need* (defined above), it holds a working herd:

$$Working\ herd = 1.5 \times Need \times (1 + 0.3 \times s)$$

where 1.5 is a buffer against a bad year and *s* is a stress ratio, the degree to which the herd exceeds the local carrying capacity.

Surplus is then drawn through two channels. The first is through secondary products:

$$S_{products} = 0.15 \times H_{productive}$$

where $H_{productive}$ is the milk- and wool-yielding part of the herd (those beyond the subsistence core), and 0.15 is a fixed yield rate, adopted for simplicity. The second channel is through culling of livestock:

$$S_{culling} = 0.35 \times 0.8 \times (H - Working\ herd)$$

Where *H* is the current herd and (*H* − *Working herd*) is the excess above what the household keeps, of which 80% is culled and each culled animal converts to 0.35 surplus

units. The culled livestock are subtracted from the herd. The year's generated surplus is the sum:

$$Yearly_{Surplus} = S_{products} + S_{culling}$$

Surplus, this model's simplified representation of capital, accumulates and decays through time. Each year it is drawn down by consumption and by loss, limiting the amount of surplus that can be accumulated. The year's balance is:

$$Total_{Surplus} = 0.7 \times Surplus_{prev} + Yearly_{Surplus} - Consumption$$

where $Surplus_{prev}$ is the previous year's holding, carried over at 0.7 reflecting an assumed 30% annual loss to spoilage and sharing; $Yearly_{Surplus}$ is the year's surplus from the herd (above); and *Consumption* is the household's yearly use of this abstract resource, set at 1.2 units per person (*Consumption* = 1.2 × *M*). In similar fashion to herd demographics, large holdings are reduced further: above 100 units a household loses an additional 5–25% a year, rising with the amount held. This represents the difficulty of storing and the social pressure to redistribute. A hard ceiling of 200 + 2.5 × *M* caps the total. Together these keep surplus a bounded, responsive measure of prosperity, the quantity that governs in part whether a household can invest in an enclosed compound (Section 5.3.3).

5.3.3 Prosperity-Based Enclosure Construction

As set out in Section 5.1, the enclosed compounds are read here not as fortresses or seats of authority but as communal infrastructure. Building one is therefore treated as a non-essential investment (Figure 5.5). A household commits part of its surplus to a permanent structure only when it is both prosperous enough to spare the resources and is in an area attractive enough to be worth anchoring to. A household that is wealthy but positioned in a poor environment, or one placed in good environment but poor in surplus and manpower, will not consider this option. The model joins these two conditions in a single prosperity index:

$$Prosperity_{inedx} = \frac{Surplus}{2 \times manpower} \times Q$$

where *Surplus* is the household's stored wealth (Section 5.3.2); and *Q* is the mean suitability of cells within ~5 km of the encampment rescaled to run from 0 (least suitable

environment) to 1 (most suitable). This equation design necessitates both surplus to manpower and environmental quality to be sufficient. A household counts as prosperous, and becomes eligible to build or reuse a compound, once this index exceeds 0.6, and only from the fifth year onward (when it has had time to accumulate surplus and attach to a broad territory). These thresholds, like the others in this section, were set during model construction as plausible working values; they stand in for a process the archaeological record cannot specify and are deliberately kept simple.

Prosperous households start by looking for an existing, unoccupied enclosed compound to reuse, since this costs far less than raising a new one. The search begins with the household's own enclosed compounds; those built within the previous 35 years. It then extends to enclosed compounds raised by other households within the last 25 years, which are regarded as a lesser choice. Reuse is possible only when the household holds at least 15 manpower and a surplus of 10, enough not to drive it into crisis (Section 5.3.4). In practice it draws just 10 manpower and 6 surplus units. A household that finds nothing to reuse turns instead to constructing one. Though construction demands markedly greater resources. New construction requires a prosperity index above 0.8, at least 30 manpower, and a surplus of 30, three times the reuse floor. It is the larger commitment, costing 20 manpower and 25 surplus units. For a new build, the household selects a location above the average environmental quality of its territory. After reuse or construction, the labourers return to the herds, and the household recovers 90% of the manpower it committed.

The effect of reusing or constructing an enclosed compound is manifested in the shared record of past occupation. That record feeds back into the suitability of the landscape and pulls households toward places already used, as set out in Section 5.3.1 for built infrastructure. An enclosed compound carries far more weight in this record than an ordinary camp. Consequently, it becomes a lasting focus of activity in the surrounding area. The household that built it is drawn back to reuse it in later years, and others may use to it as well, though this is less probable. This mechanism leads activity to gather around standing enclosed compounds, further affecting the palimpsest pattern (Section 5.2.4).

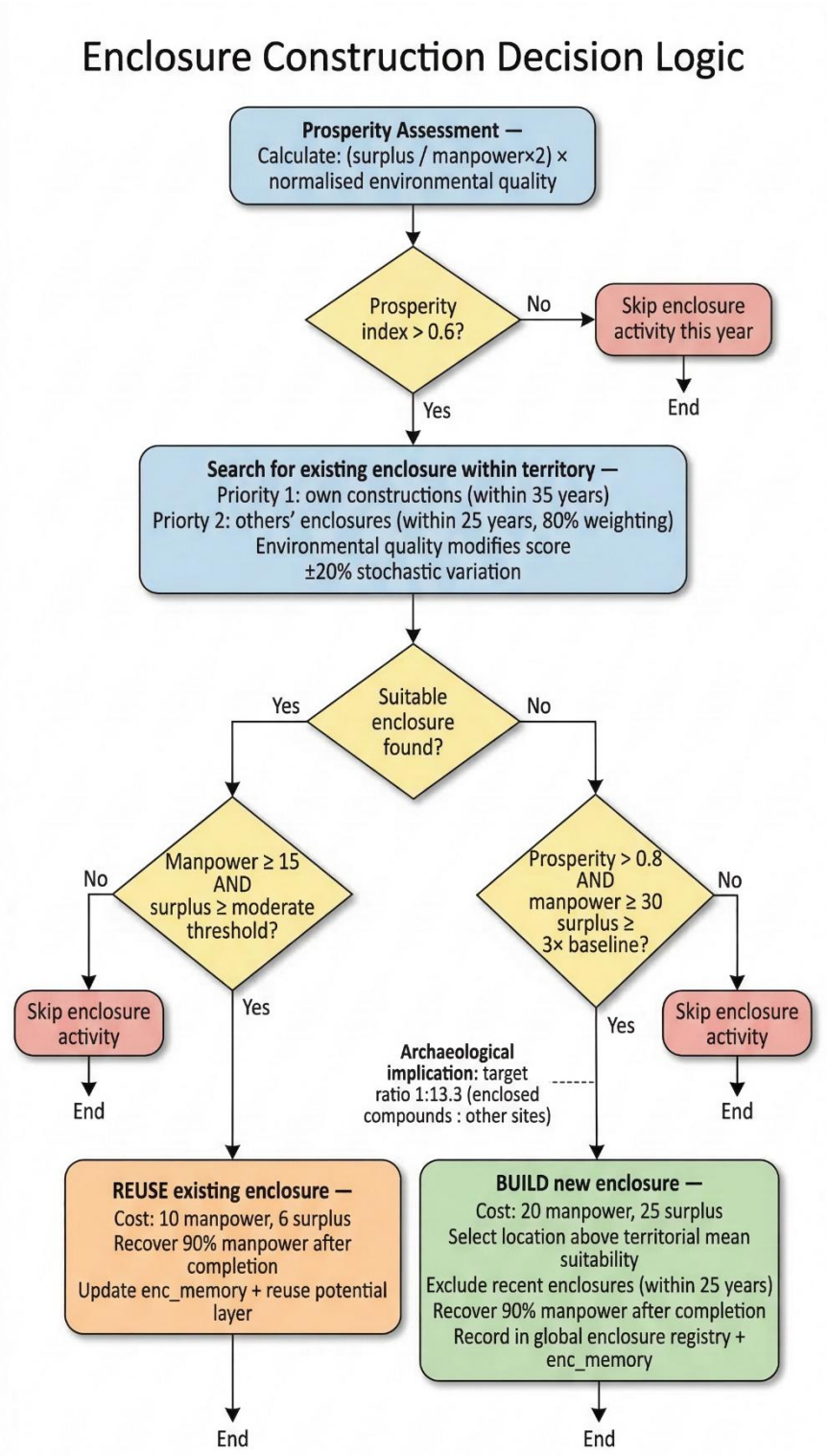


Figure 5.5. Enclosure construction decision logic. Prosperity index gates all enclosure activity at threshold >0.6. Qualifying households first search for reusable enclosures, prioritising own constructions within 35 years, then others' within 25. Reuse requires manpower ≥15 and moderate surplus (cost: 10 manpower, 6 surplus). New construction requires prosperity >0.8, manpower ≥30, and surplus ≥3× baseline (cost: 20 manpower, 25 surplus).

5.3.4 Crisis Response and Survival Strategies

When a household's resources run low, it enters a crisis. The surplus it has built up in better years, its accumulated capital from herding, farming, and trade (Section 5.3.2), serves as the reserve it now draws on. Some years empty that reserve faster than the agents can refill it. When rainfall is poor the pasture contracts and the herd stop growing (Section 5.3.2), so little new surplus is produced, while consumption and storage losses continue to exhaust the reserve. A run of such years pushes the household into a survival crisis. The model triggers the crisis when surplus falls below 10 units, an tentative floor beneath which a household can no longer sustain itself within the model's simplified system. To rebuild the reserve the household turns to its herd, selling animals to bring it back above the floor.

The herd is the household's only liquid asset in the model; thus, it can be converted into surplus and back at a fixed rate of 0.3 units per animal. During crisis, the household can choose to sell livestock to close the gap between its surplus and the 10-unit threshold. This 0.3 return sits below the 0.35 a household realises from orderly culling in a normal year (Section 5.3.2), reflecting a sale under pressure. This selling is limited by a minimum reserve of 18 animals per person, so a household does not fall below its subsistence needs (Section 5.3.2). The reverse conversion may occur in easier times. A household holding more than 80 surplus units with a herd still relatively small (below 600 livestock) can convert part of its capital to livestock, spending up to 20% of its surplus (at most 40 units). While this is not a crisis response, it is its mirror image: a secure household rebuilding productive capacity rather than consuming all the herd down to survive.

When a household cannot meet the survival threshold, it fails. The survival threshold requires a household to have a surplus of zero or more, a herd of at least 15 animals, and manpower above zero. If any of these criteria is not met, the household is no longer functional and is removed from the simulation. Whatever manpower, surplus, and livestock it holds pass to a new replacement household, formed at the end of the year. This new household starts with a fresh herd of about 900 goats on top of the inherited remainder, representing in a very simplified way the remnants of an expired household seeding a new one. Replacement is one-for-one, so the number of households stays constant across the run, in keeping with the model's constant-population assumption

(Section 5.1). This turnover may be uneven throughout the simulation, and can cluster after runs of poor years that push several households into crisis at once. The households that endure, meanwhile, rebuild through ordinary herd growth, which is fastest when a herd is small (Section 5.3.2), so the surviving population recovers as conditions ease.

5.4 The Simulation Run

The previous two sections set out the model's components (Section 5.2) and the decisions and processes that advance it (Section 5.3); here I describe how they run together. A simulation run begins by setting the initial conditions (Section 5.2.4), advancing a year at a time, repeating the annual cycle for a set number of years, 75 in the calibration and validation runs (Figure 5.6). Each year the households act in random order (as not to give advantage to any specific household), and the environment is then updated to set up the next year. Over the full run of 75 years, the occupation pattern emerges from the interaction between the households and their environment (Section 5.1).

Within a year cycle (Figure 5.3) the households act one after another, in a random order. First, each household selects a camp, sets its territory and disperses the nuclear families within it (Section 5.3.1); then it works through the following behaviours and processes: the herd updates (grows or contracts) and surplus is calculated (Section 5.3.2); then it decides whether to reuse or build an enclosed compound (Section 5.3.3); and, should its reserve fall short, it responds to crisis (Section 5.3.4). The ecological and archaeological imprint each household leaves is recorded for the environmental accounting that follows (Figure 5.7).

The transition to a new year starts by the model updating the environment. First, it draws a fresh annual rainfall from the record and recomputes the resources that depend on it, then it folds the year's occupation into the accumulating degradation and reuse potential, and recalculates each cell's suitability. This hands off to a new yearly cycle (Figure 5.3). The cycle repeats until the final year, which closes without a further transition.

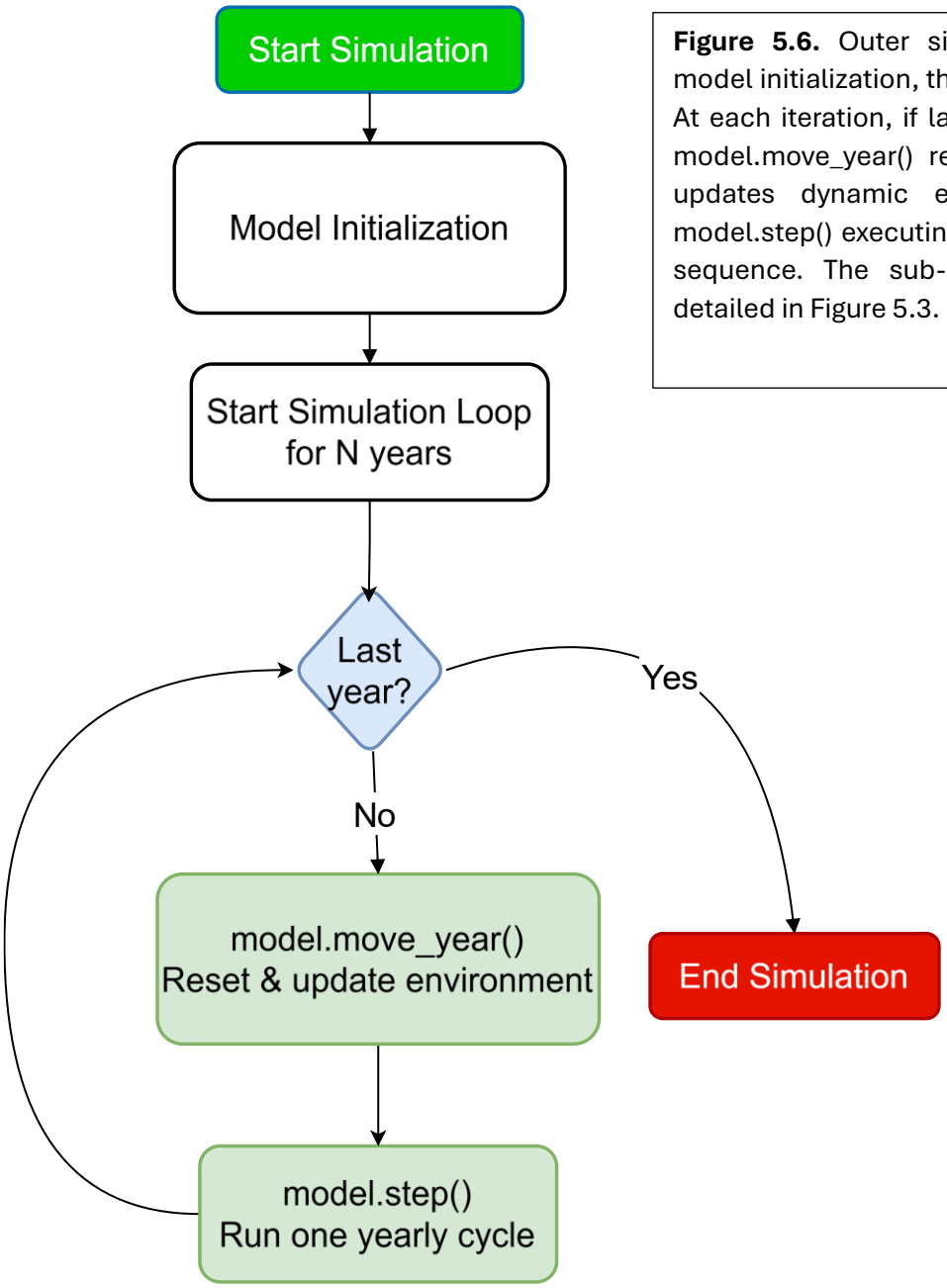


Figure 5.6. Outer simulation loop structure. Following model initialization, the simulation iterates for N (75) years. At each iteration, if last year, the simulation ends, if else, model.move_year() resets household grid positions and updates dynamic environmental layers, followed by model.step() executing the full annual household decision sequence. The sub-processes within model.step() are detailed in Figure 5.3.

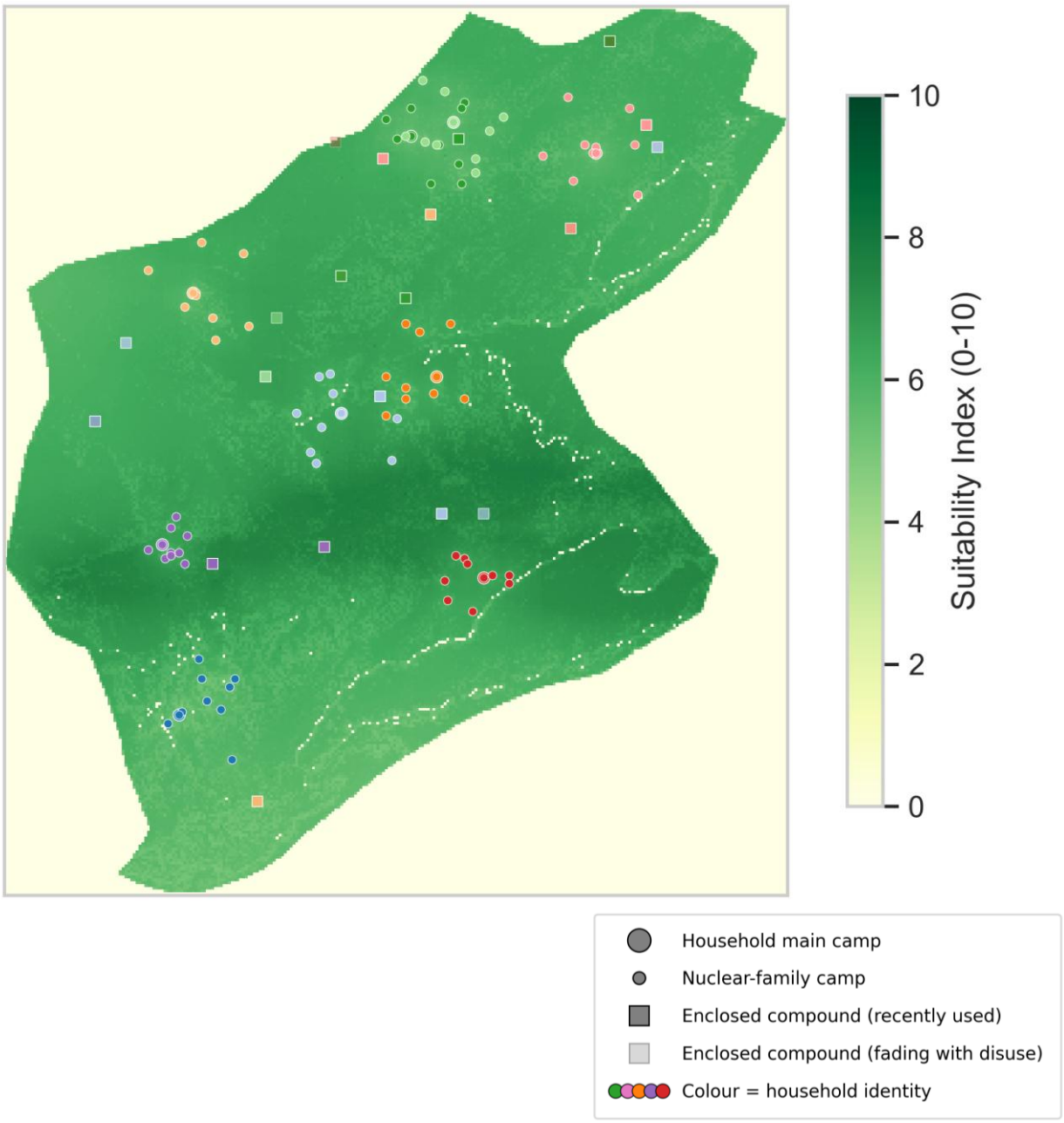


Figure 5.7. Snapshot of a single simulated year (Year 25 of a 75-year run). The green surface is the composite environmental suitability index (0–10; Table 5.1) that each household reads when choosing where to camp. Filled circles mark camps, coloured by household: the larger circle in each cluster is the household's main camp and the smaller circles its nuclear-family camps (§5.3.1). Squares mark enclosed compounds, coloured by the household that last used them and fading as they fall out of use (§5.3.3).

5.5 Model Calibration and Validation

This section details the calibration and validation of the model against the archaeological record. Calibration, in this context, is defined as narrowing the model's free parameters to a plausible range, using an external data source, rather than assuming them (Romanowska et al., 2021). The seven environmental weights (Section 5.2.1.2) are varied until the combination is found whose simulated record best reproduces the observed Iron Age IIA record, according to two independent patterns (described below). Validation then tests how closely that best combination reproduces those patterns.

5.5.1 Calibration Strategy and Rationale

The model, following Pattern-Oriented Modelling (POM) framework, employs a multi-criteria calibration process especially fit for agent-based models where emergent system-level spatial patterns form the target of calibration (Grimm et al., 2005; Gallagher et al., 2021). This approach addresses a fundamental challenge in demonstrating the plausibility of the palimpsest model: showing that the simulated household decisions about camp placement, resource management, and enclosure construction drive the emergence of the archaeological patterns observed in the Iron Age IIA Negev Highlands.

Calibrating a pattern-oriented model requires identifying parameter combinations that enable the above-described processes and behaviours to simultaneously reproduce both the regional spatial pattern (regardless of site types) and the enclosed compounds to other sites ratio (henceforth site-type ratio). This multi-pattern matching is aimed at reducing equifinality - while many parameter sets might match either the regional spatial pattern or the site-type ratio, far fewer also generate both. Accordingly, calibration aspires to find parameter values combination that lead to similar patterns consistently.

Based on the rational above, the calibration examined the similarity of the model results to the real archaeological record through the following evaluation functions: (i) spatial distribution of all sites measured through standard deviational ellipse overlap; and (ii) site-type ratios (both are defined and quantified in Section 5.5.3). These patterns were selected for their diagnostic power while avoiding metrics that score the prediction of exact site locations (such as Root Mean Square Error, common in other models). The standard deviational ellipse captures where activity concentrates and along what axis it

spreads. This allows the two distributions to be compared at the regional scale without penalising the model for the location of any individual site. The site-type ratio addresses a key prediction of the model, that enclosed compounds are an expression of accumulated surplus rather than a functionally separate category (Section 5.1). It therefore tests whether the model's decision-making and enclosure construction and reuse rules (Section 5.3.3) generate them at frequencies resembling the observed ones.

5.5.2 The Calibration Procedure

Calibration focused exclusively on the seven environmental suitability weights deliniated in Section 5.2.1.2. These weights determine how strongly each environmental variable influences household behaviour, making them the primary drivers of the emergent spatial patterns. Other model parameters remained fixed based on archaeological evidence, ethnographic analogues, or practical considerations of model development as detailed in sections 5.2 and 5.3. Fixing these parameters focuses calibration on the environmental criteria while acknowledging that multiple aspects of household behaviour rest on user defined values independent of the specific Negev Highlands environment.

Spatial similarity is evaluated by the overlap of standard deviational ellipses fitted to the simulated and observed site distributions. A standard deviational ellipse summarises a scatter of points by an ellipse centred on their mean, with its two perpendicular axes scaled to the spread of the points (here, two standard deviations) along the directions of greatest and least dispersion; its size, elongation, and orientation together describe where points concentrate and along what axis they are drawn out (Yuill, 1971). Comparing the two ellipses therefore compares the spread and orientation of the two distributions directly, without assuming the sites follow any particular statistical form. The overlap of the two ellipses is quantified by the Intersection over Union (IoU):

$$IoU = \frac{A_{overlap}}{A_{overlap} + A_{symmetric\ difference}}$$

Where $A_{overlap}$ represents overlapping area and $A_{symmetric\ difference}$ represents non-overlapping area. The spatial error component is calculated as: $spatial_{error} = 1 - IoU$, ranging from 0 (perfect overlap) to 1 (no overlap).

Site-type similarity is the absolute difference between the simulated and observed proportions of enclosed compounds among all sites:

$$ratio_{error} = \left| p_{archaeological} - p_{simulated} \right|$$

where p is the site-type proportion. Like the spatial error, it ranges from 0 (identical proportions) to 1 (completely different).

Each simulation run was scored by a composite objective function that combines the two pattern-similarity measures described above, weighted equally to reflect their complementary diagnostic value:

$$total_{error} = \frac{(spatial_{error} + ratio_{error})}{2}$$

This $total_{error}$ is what the calibration seeks to minimise. To account for stochasticity in household decisions and yearly rainfall, each weight combination is run for ten replicate simulations with different random seeds. The ten scores are then averaged, with the mean taken as that combination's score, so a good result reflects consistent performance rather than a single fortunate run.

Parameter calibration used Optuna (Akiba et al., 2019), a hyperparameter-optimisation framework — the equivalent, in machine learning, of calibration — that efficiently searches large parameter spaces through adaptive sampling. The parameter space is the set of all possible weight combinations, with one axis per calibrated parameter (seven in this case); each axis may be continuous or, as here, discrete. As set out in Section 5.2.1.2, each weight multiplies its corresponding environmental variable, and the weighted variables are summed to form the suitability raster.

Each of the seven weights was set as a discrete integer from 0 to 7, giving eight ordinal levels of importance per variable. This discretisation serves two purposes. First, it keeps the search space finite: seven continuous weights would span infinitely many combinations, whereas seven weights with eight levels each define a large but finite space of 8^7, i.e., ≈ 2.1 million combinations. Second, the ordinal scale gives an intuitive reading of importance, where 0 excludes a variable, low values (1–2) mark minor influence, moderate values (3–5) intermediate importance, and high values (6–7) a

dominant influence. Because the weights express relative importance, they are normalised to sum to 1 before use.

The calibration used two of Optuna's built-in mechanisms to save computing time. First, a sampler chooses which weight combinations to try, learning from the trials already run and concentrating the search where good scores have already been found (Tree-structured Parzen Estimator, TPE; Watanabe, 2023). Second, a pruner abandons a trial early once its intermediate scores fall behind the better trials so far (MedianPruner; technical details in Appendix 5.3). Formal calibration ran 200 such trials, each comprising ten replicates, preceded by exploratory manual runs used to check that the model behaved sensibly.

Calibration stopped after the predetermined 200 trials rather than on a convergence criterion, a fixed-trial approach driven mainly by computing constraints. The global optimum may therefore remain undiscovered, given the size of the parameter space, an issue revisited in Section 5.5.4.

5.5.3 Target and Simulated Patterns

The observed and simulated records are not symmetrical, a fact that directly shapes what counts as a good match. The archaeological record is an accumulation of seasonal activity across the period, further filtered by differential preservation, variable survey coverage, and post-depositional processes. Some sites may remain undiscovered, though the overall survey coverage (Chapter 2) is taken to be representative. The simulated record, in contrast, is complete, holding every household position and enclosed compound over the run. Calibration therefore aims not for an exact match but for a simulated pattern close enough that the remaining differences could plausibly be attributed to this asymmetry.

The simulated record is turned into comparable patterns in two steps. First, each cell accumulates an occupation signal over the run: a household main camp adds 0.5 to its cell each year it is used, and a dispersed nuclear-family camp adds 0.25. Summed across the 75 years, a cell counts as a site once its signal reaches 0.6 (a threshold set by face validation), which takes at least two seasons at a main camp or three nuclear-family stays. A single one-off visit falls below the line, filtering the most ephemeral activity while

retaining what would plausibly leave an archaeological trace. Above the threshold a cell counts as a site whether it was occupied just enough to qualify or intensively for decades, mirroring how the archaeological record is read: a site is recorded by the presence of stone-built structures, regardless of their extent and longevity. Second, every cell where an enclosed compound was built is additionally marked as an enclosed-compound site, identified from the model's construction list. The spatial distribution uses all sites together, while the site-type ratio is the number of enclosed-compound sites relative to the number of other sites.

Calibration targets:

- **Spatial distribution.** The observed spatial distribution's standard deviational ellipse covers 1,585.02 km^2, with a centroid at 168,369.37 E, 515,592.33 N (ITM grid; Figures 5.1, 5.8), a main axis of 66.68 km and a perpendicular minor axis of 30.26 km, an axis ratio of 2.2:1. This elongation reflects the broadly east-west running concentration of Iron Age IIA activity along the line from Tell el-Qudeirat through Ramat Barnea and Ramat Matred to Avdat, diverted slightly northwards (55.66°) due to the presence of sites also in the northeastern parts of the region. This pattern, as already discussed in Chapter 3, correlates with the hypothesised movement corridor toward Faynan in the northern Arabah. The mean centre and the median centre (168,223.20 E, 512,622.45 N) differ by about 3 km toward the south-southwest, indicating clustering of sites in the southern part of the ellipse. This pattern is the calibration goal, which the simulated distribution is meant to replicate.
- **Site-type ratio.** Of the 462 sites analysed within model boundaries (Section 5.2.1.1), 32 (6.93%) contain enclosed compounds. The remaining 430 (93.07%) have other architectural forms, including circular structures, rectangular buildings, towers, and four-room houses, singly or in combination; following Chapter 3, five sites with questionable enclosures are counted here among these non-enclosure sites. The observed ratio is therefore about 1:13.4 (enclosed compounds to other sites), the second calibration goal.

5.5.4 Validation Results: Model-Data Comparison

Of the 200 calibration trials, 143 completed all ten replicates. The remaining 57 were terminated early by the MedianPruner because their intermediate objective-function values were unlikely to improve upon the current best score. All subsequent results derive from the 143 completed trials. During this calibration phase, the highest-performing weight combination achieved a composite objective value of 0.2013. Its specific weight structure is illustrated in Figure 5.9 and is examined alongside the other leading trials below.

To confirm that this result is not an artefact of a single fortunate set of random seeds, the best-scoring combination was re-run for 30 additional replicates. Across these replicates it returned a mean composite score of 0.213 ± 0.021, within 0.011 of the 0.2013 calibration score, confirming that the combination performs consistently rather than depending on a lucky seed. How closely it reproduces each of the two calibration goals, the spatial distribution of sites and the site-type ratio, is set out below (mean spatial error 0.314 ± 0.039; mean site-type error 0.111 ± 0.021).

Comparing the simulated spatial distribution with the observed one shows close agreement in centre and overall size, with the remaining difference lying in shape. The standard deviational ellipses of the observed Iron Age IIA site distribution and of the best-performing trial (Figure 5.8) overlap substantially across the central Negev Highlands (IoU 0.68); their centroids nearly coincide (about 1.9 km east and 0.8 km south), and their areas are almost equal (1,602.43 km^2 simulated against 1,585.02 km^2 observed, Section 5.5.3). The residual lies in orientation and elongation rather than position or size: the observed ellipse is strongly drawn out along the ENE corridor axis (ratio 2.2:1), whereas the simulated one is markedly rounder and less strongly oriented (ratio 1.25:1). The modelled households thus spread more evenly around the core, reaching less far along the corridor and further across it than the surviving record does. This shape mismatch, not any difference in size, drives the bulk of the spatial error. It is consistent with the observed–simulated asymmetry noted above, though it may also reflect the model's own simplifications, further discussed in Section 5.7.

Site-type similarity, the second calibration target, is reproduced less closely than the spatial pattern. In this run the best combination overproduces enclosed compounds: they make up about 17.6% of simulated sites against the observed 6.93% (Section 5.5.3), a simulated ratio near 1:5 versus the observed 1:13.4, roughly 2.4 times as many. This overproduction may reflect prosperity thresholds that admit enclosures more freely than the actual system did, or a site-classification threshold that inflates the count of classified sites.

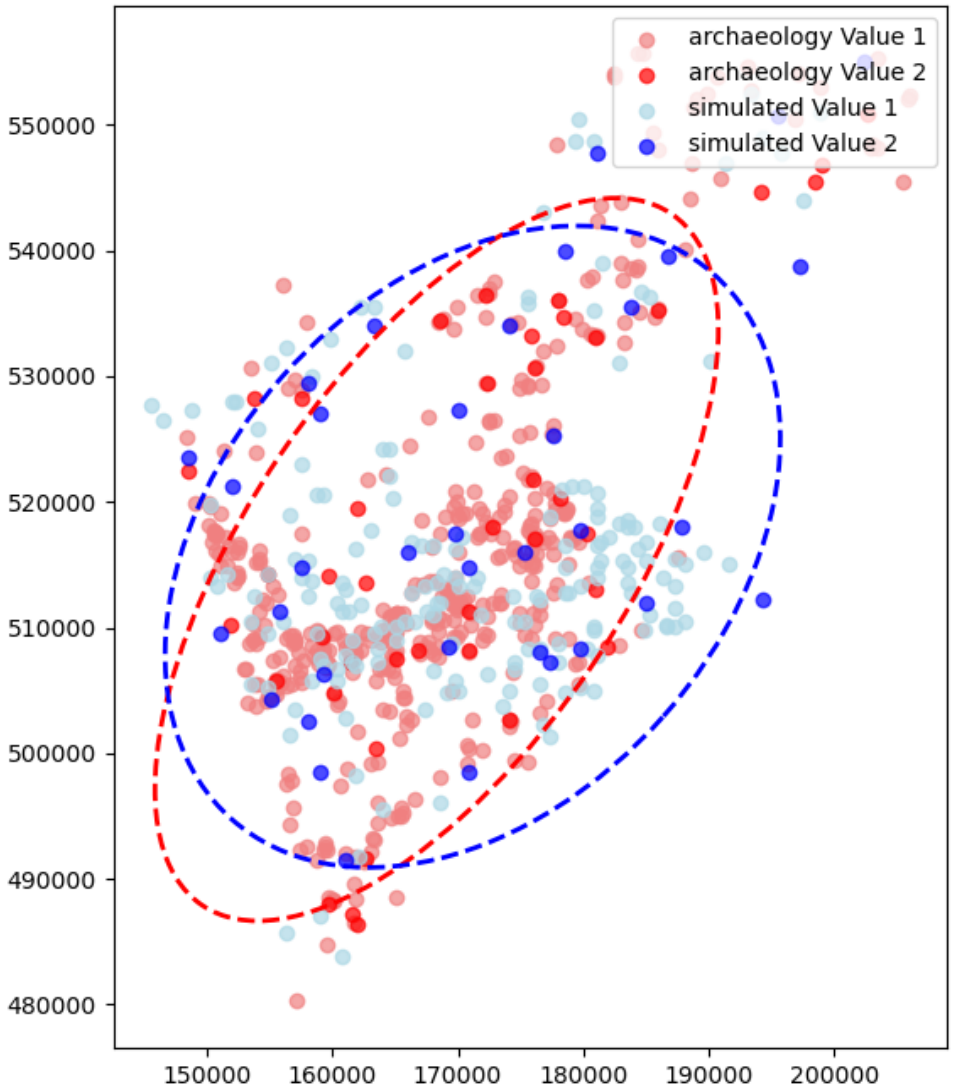


Figure 5.8. Spatial distribution comparison between observed and simulated Iron Age IIA sites. Red points: archaeological sites (light = other sites, dark = enclosed compounds); blue points: simulated sites from best performing. Dashed ellipses represent standard deviational ellipses at two standard deviations.

To place the best trial in a broader context and gauge how unique its parameter structure is, the leading cluster of trials was examined together: the 19 trials scoring below 0.21 on the composite objective, all within a hundredth of the best. Across this cluster the weight structure of the best trial proves stable rather than idiosyncratic. Proximity to the Tell el-Qudeirat–Faynan corridor and reuse potential consistently carry the highest weights (mean 0.304 ± 0.005 each), followed by multi-annual resource exhaustion (0.217 ± 0.010), the region's long-term mean rainfall (0.132 ± 0.009), and proximity to permanent water (0.041 ± 0.010). Slope and the current year's rainfall receive zero weight across all 19 trials without exception. The very narrow standard deviations mark this weight structure as a stable attractor in the parameter space, not a single fortunate draw. This substantially reduces concern over equifinality for the dominant variables, though the exact proportions within the cluster cannot be pinned down uniquely.

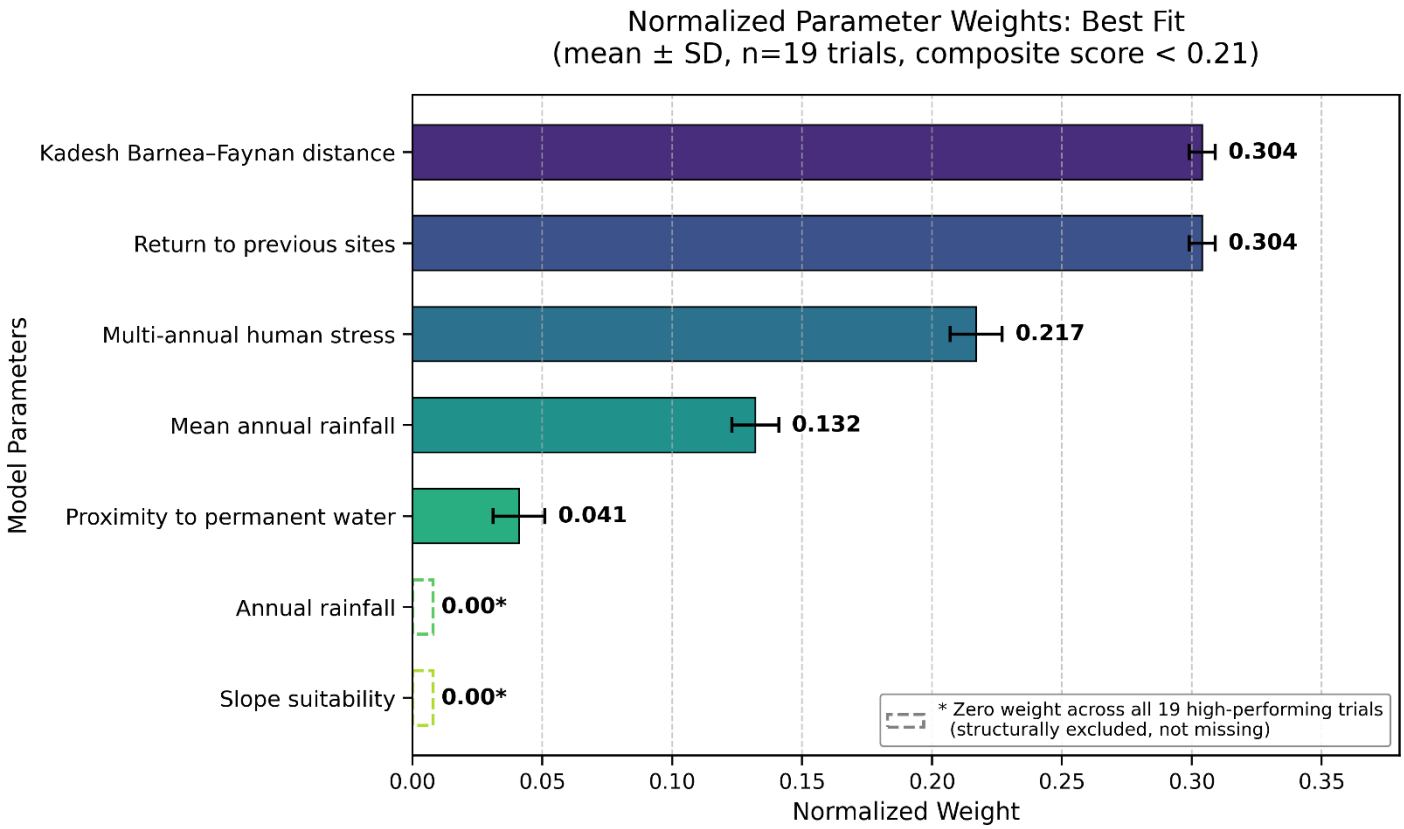


Figure 5.9. Normalised mean parameter weights across 19 high-performing trials (composite score <0.21), with error bars showing ±1 SD. Proximity to the Tell el-Qudeirat–Faynan corridor and reuse potential co-dominate (0.304 ± 0.005 each), followed by multi-annual resource exhaustion (0.217 ± 0.010), mean annual rainfall (0.132 ± 0.009), and proximity to permanent water (0.041 ± 0.010). Annual rainfall and slope suitability receive zero weight across all 19 trials (dashed outlines) (§5.5.4).

5.6 Sensitivity Analysis

The calibration in Section 5.5 tuned only the seven environmental weights; every other setting of the model — how long it runs, how many households it starts with, how territory is handled, and where the cut-off for counting a cell as a site is placed — was fixed beforehand on archaeological, ethnographic, or practical grounds (Sections 5.2–5.3). Sensitivity analysis asks how far the calibrated result depends on those fixed choices, and whether it genuinely reflects the modelled behaviour rather than the settings surrounding it (Romanowska et al., 2021). First, a null-weight baseline tests whether the calibrated weights achieve statistically better results than equal weighting. Afterwards, holding the calibrated weights constant, four choices (out of many) are examined: (1) the temporal horizon, testing whether the 75-year run length is a stable stopping point; (2) the number of household agents; (3) territorial mobility, comparing the behaviour described in Section 5.3.1 against households confined to a fixed territory; and (4) the site-classification threshold, testing how the number and spread of simulated sites respond to where that cut-off is set. All tests except the null baseline use the calibrated weights of the best-performing trial (Section 5.5.4) and are scored with the same metrics. Full methods, tables, and figures appear in Appendix 5.7; this section reports the principal findings.

Null Weight Baseline

The point of this baseline is to check that the calibrated weights perform better than arbitrary ones. If giving every variable equal importance matched the calibrated result, the optimisation would have found nothing worth keeping. Thirty iterations with all seven weights set equal (1/7 ≈ 0.143 each) yielded a mean composite error of 0.318 (95% confidence interval 0.309–0.328). The best calibrated trial, at 0.201, falls entirely below that distribution with a 36.7% reduction in error (Figure A5.1), confirming that the calibrated weight structure captures a genuine signal that this baseline cannot reproduce.

Temporal Horizon and Population Size

These two tests ask whether the 75-year, nine-agent baseline was set too conservatively — whether running the simulation longer, or starting it with more households, would

reproduce the record more closely. The length of the simulation was tested by extending a single run to 150 years and evaluating it at ten-year checkpoints. The population was tested by running five, fifteen, and twenty households against the nine-household baseline, five replicates each. On the composite score, the longer horizon and the larger populations both appear to help, each lowering the error relative to the calibration results (Section 5.5.4). Yet a visual inspection raises the concern that the improvement stems from the way the patterns are evaluated. At longer durations and higher agent counts, simulated activity spreads until it covers large parts of the region (Appendix 5.7, Figures A5.2–A5.5), a distribution that is not realistic whatever its score. The cause, I suspect, lies in the spatial distribution metric: ellipse overlap does not account for the number of sites produced, so the more cells that register activity across the area of the observed ellipse, the greater the overlap and the better the score. The better scores at 150 years and at 15–20 households therefore cannot be read as genuine improvements. Reducing the population to five households failed from the opposite direction, generating too few sites to match the regional spread. Therefore, the nine-agent, 75-year configuration remains a sound choice. These results suggest that the demographics and activity intensity, held constant across all years, may be oversimplified; whereas demographic and behavioural patterns are likely to change over the period, and incorporating those aspects into the model, would be a natural direction for future work.

Territory Flexibility

This test isolates the contribution of the region-wide repositioning described in Section 5.3.1. In this fixed-territory mode, households kept the territory they were first assigned and could move freely only within it. The two modes scored close on average (mean error ~0.25 fixed versus ~0.21 flexible), which is unsurprising, since the fixed mode removes only region-wide moves while retaining small-scale movement inside the territory. The clearer difference is in consistency; the fixed mode was far more variable, with one run reaching 0.41 (Figure A5.6). That spread reflects its dependence on the initial placement — with no region-wide repositioning, a household located in poor ground cannot relocate to seek better, and remains confined to ground it would otherwise have abandoned. Region-wide repositioning, by contrast, lets households respond each year to an environment degraded by accumulated occupation or thinned by a poor rainfall year,

which yields a lower and more stable score. The mechanism is thus best understood as a stabilising one.

Site-Classification Threshold

The final test concerns the 0.6 threshold that turns the simulated occupation surface into discrete sites (Section 5.5.3). Because that value was set by face validation rather than derived, the threshold was varied from 0.1 to 1.0, on a single run's output, to see how far the two calibration targets depend on it. The number of sites proved highly sensitive to it: the 245 sites at 0.6 rise to 1,236 at 0.3–0.5 and to nearly 6,000 at 0.1–0.2 as more ephemeral visited cells are counted; it falls to just 73 at 0.8–1.0. The site-type ratio was affected markedly as well, since enclosed compounds are counted from a construction list, regardless of occupation intensity, while every other site must clear the set threshold. As the threshold rises and more cells drop away, the enclosed compound share therefore climbs: from under 1% at the lowest levels, through 16.7% at 0.6, to 56% at the highest. The observed share of 6.93% is not replicated at any level of this discretisation, falling between the 0.5 (3.3%) and 0.6 (16.7%) settings. The underproduction of 'other' sites seen in the calibration (Section 5.5.4) thus stems partly from this threshold. Although the spatial overlap improves at lower thresholds, it does so only by inflating the sites to archaeologically implausible levels, as with the temporal horizon and population tests above. A more refined classification rule, for example one that counts an ordinary site on the same evidential footing as an enclosed compound, where recurrent occupation implies a surviving built feature rather than a summed-intensity value merely exceeding a threshold, is a clear direction for future work.

5.7 Summary and Discussion

At the start of this chapter (Section 5.1) I asked whether the local, everyday decisions of mobile pastoral households could, without top-down direction, generate a record like that of the Iron Age IIA Negev Highlands. The simulated activity under the calibrated weights gathers in the same parts of the region as the actual sites, at a comparable centre and extent, within the fixed placement area the model is allowed to occupy (Section 5.5.4). This distribution emerges from the households' operationalized rules, reflecting

the bottom-up processes the palimpsest model hypothesises. The sensitivity analysis points in the same way. Region-wide repositioning gave a lower and more consistent fit than confining households to fixed territories, so the annual movement at the core of the palimpsest reading is what stabilises the pattern (Section 5.6). Longer runs and larger populations did not reproduce the record more closely; they only spread activity more widely across the region (Section 5.6). The nine-household, 75-year configuration is therefore a sound basis for the sufficiency claim, though some limitations and room for improvement do exist.

The cumulative, region-wide palimpsest is built into the model through the households' yearly choice of where to camp (Section 5.3.1), yet the way it accumulates shows a few informative trends. In a sample run, enclosure building proved to come fast: 26 of the 41 compounds built over the run are erected within the first 25 years. Once the region reaches a relative saturation of such structures, and as reuse costs less than building anew, only five new compounds appear in the final 35 years (Figure 5.10). The 41 compounds accumulate 519 visits between them, a mean of 12.7 each and 44 at the most reused. Early investment thus becomes a lasting attraction, reinforced with each return. The same trajectory is observed in ordinary occupation. Most cells that were used hold only one or two visits and never reach the threshold that counts a cell as a site (Section 5.5.3); fewer cells cross this threshold, and of those only a handful are revisited far above it (Figure 5.11). Hence, the simulated record can be seen as reflecting a range of human activity, from ephemeral traces to a few repeatedly visited cores, which can be correlated to some extent with the intra-site variation documented in Chapter 4 (and see further discussion in Chapter 6).

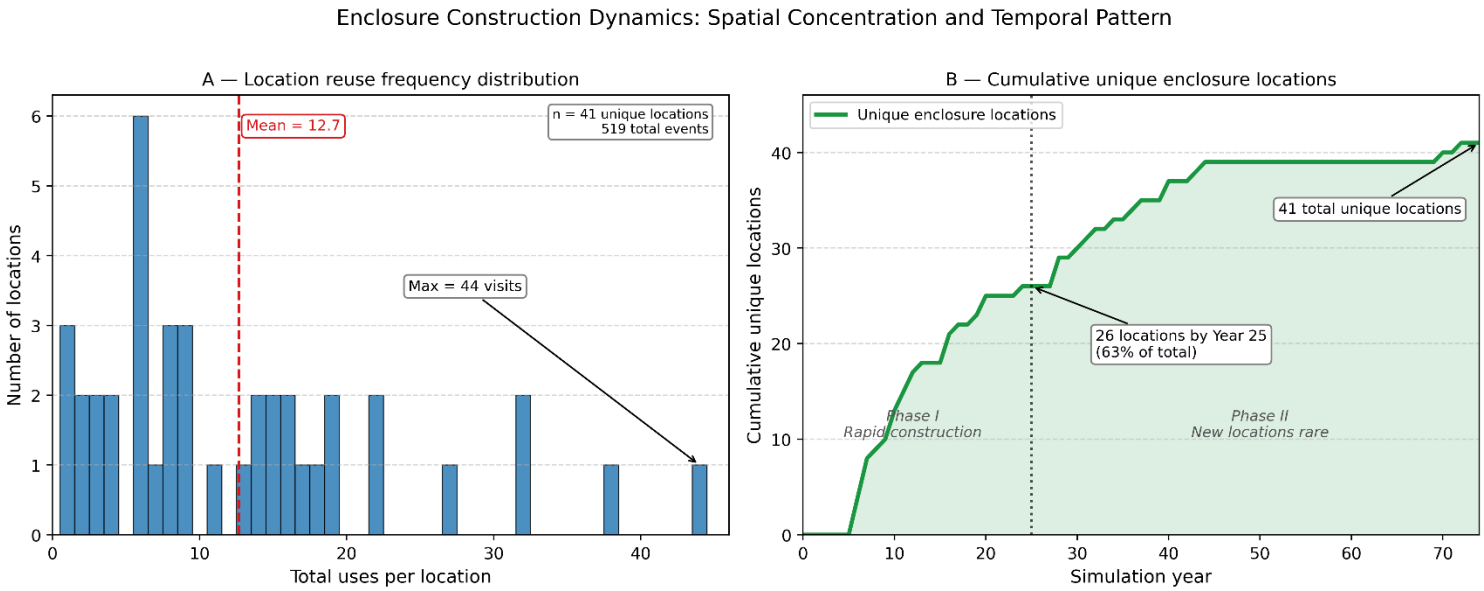


Figure 5.10. Enclosure construction dynamics across 75 simulation years of a sample run. **A**: location reuse frequency distribution across 41 unique enclosure locations (519 total construction and reuse events); mean 12.7 uses per location (red dashed line), maximum 44 visits at a single location. **B**: cumulative unique enclosure locations by simulation year, showing two distinct phases: Phase I (Years 0–25), rapid construction establishing 26 locations (63% of total); Phase II (Years 25–74), reuse dominates with only 15 new locations added.

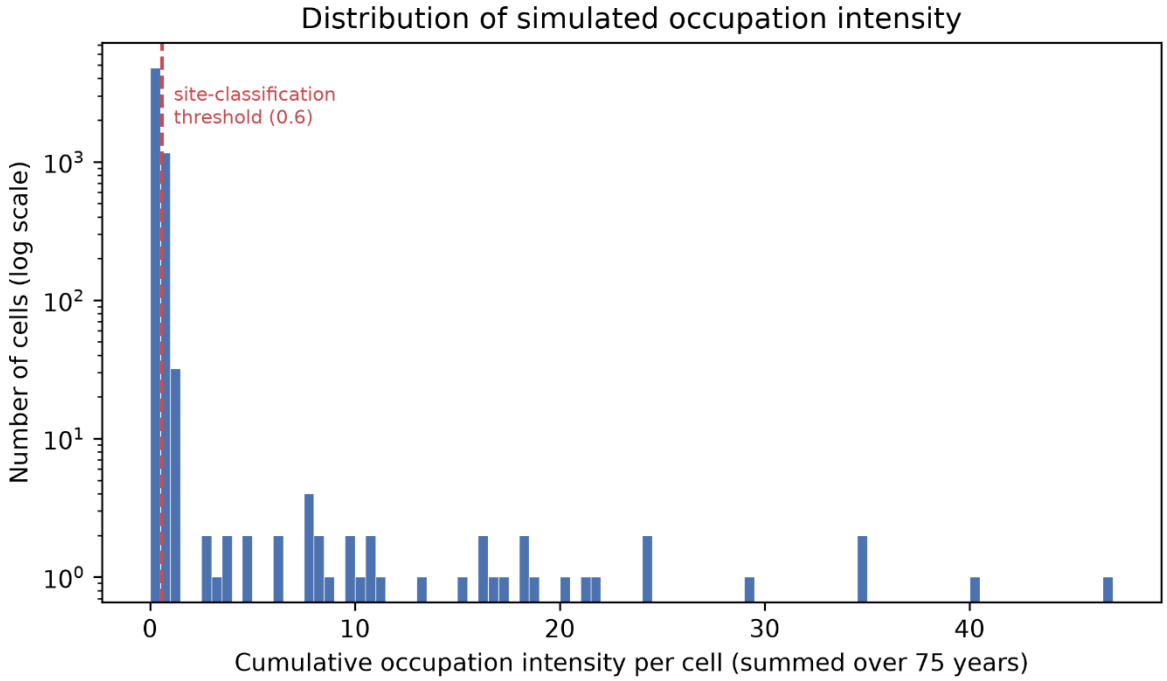


Figure 5.11. Distribution of simulated cumulative occupation intensity per cell across the 75-year run (best-trial seed-173 output; log-scaled count axis). The dashed line marks the 0.6 site-classification threshold (Section 5.5.3).

The calibration identified proximity to the Tell el-Qudeirat–Faynan corridor and reuse potential as the two most influential variables (Section 5.5.4). The convergence of the point-pattern analysis (Chapter 3) and the calibrated ABM, both assigning corridor proximity high importance, is telling, and is further discussed in Chapter 6. The substantial weight also given to reuse potential is best understood, as discussed above, as a kind of positive feedback loop, in which repeated visits and standing structures weighed heavily on households' decisions, perhaps reflecting both traditional locational preferences as well as the energy saved by reusing existing infrastructure.

The low weights bear on the interpretation as much as the high ones. Proximity to permanent water fell to near zero, matching Chapter 3's finding of no significant attraction to permanent water sources ($p = 0.619$) and pointing again to mobile groups not tied to fixed springs. Slope also received no weight, for either of two reasons: it may parallel the limited relevance of slope found in Chapter 3, or it may be structural, i.e., the steepest cells, those scoring below 1 on the fuzzy slope suitability, being excluded before the weighted summation, so that the slopes which are left varies little. The current year's rainfall likewise received zero weight, read here as a limitation of how rainfall was represented rather than as evidence that rainfall did not matter. The modern rainfall record was applied in random yearly order, which most plausibly does not resemble the sequences of the Iron Age IIA closely enough to have carried real influence. A further possibility is that the derivatives of rainfall, the pasture and cultivation it drives, would have been more informative terms to calibrate than rainfall itself.

The model falls short in a few aspects. The main gap between the simulated record and the observed one is the shape of the simulated distribution. The model spreads activity too evenly around the region, producing a rounder ellipse than the corridor-aligned elongated one (Section 5.5.4). It also builds enclosed compounds too readily, at roughly 2.4 times higher than their observed share, and produces correspondingly few other sites. The sensitivity analysis shows that part of this imbalance follows from the site-classification threshold rather than from the building behaviour (Section 5.6), hinting at a possible avenue for future improvement. Nonetheless, the calibrated model presents a pattern closely reproducing the target one (i.e., the ‘real’ archaeological record). It shows that the modelled local behaviours are a plausible way to generate this record, though

other alternatives remain open. Tightening the site-type ratio, penalising over-dispersion in the objective function, and reconsidering how annual rainfall would be most effective constitute the clearest routes to improvement.

Feature Type	Rekhes Halukim	Nahal Lavan	Nahal Mitnan	Nahal Resisim	Zin 231	Total
Small Structure (1.5-3m)	0	7	11	3	8	29
Large Structure (3-10m)	6	15	10	7	3	41
enclosure (>10m)	0	5	1	2	9	17
Storage Silo	4	0	0	0	0*	4
Wall	2	7	6	5	3	23
Installation	0	7	5	7	5	24
Cairn	1	0	4	2	0	7
Stone Concentration	0	3	3	2	3	11
Cupmark	0	3	0	0	0	3
Terrace	0	1	0	0	0	1
Other	0	0	1**	0	1***	2
TOTAL	**13**	**48**	**41**	**28**	**32**	**162**

Table 4.3. Immobile Finds Inventory by Locality.

* Zin 231 includes probable storage silos documented as "small structures" due to lack of excavated verification.
** *Nahal Mitnan "Other" constitutes 1 Stone Scatter.*
*** *Zin 231 "Other" constitutes 1 Cleared Surface.*

Temporal\Origin	Environmental	Anthropogenic
Permanent	Mean annual rainfall (✓) Slope (✓) Distance to permanent water (✓) Proximity to arable soils (✗) Vegetation type (✗)	Distance to Tell el-Qudeirat–Faynan corridor (✓)
Annual/Dynamic	Annual rainfall (✓) Available pasture (✗) Fertile arable soils (seasonal) (✗)	Multi-annual resource exhaustion (✓) Reuse potential (✓)

Table 5.1. Environmental variables classification system. Variables are organised along two axes: temporal (Permanent vs. Annual) and origin (Environmental vs. Anthropogenic). Checkmarks (✓) indicate variables included in the weighted suitability summation; crosses (✗) indicate variables retained for resource calculations only, excluded from direct summation. Full transformation specifications for each variable are provided in Appendix 5.1.

Appendices

Appendices are numbered according to their related chapter number. Appendices 1 and 4 are long, big excel tables and hence are not included here. They are available upon request from the author (itay.lubel@mail.huji.ac.il)

Appendix 5: Model Technical Specification

A5.1 Environmental Parameter Specifications

All environmental parameters were normalised to a 0–10 suitability scale through fuzzy membership transformations (ArcGIS Pro Spatial Analyst) prior to weighted summation. Distance-based parameters were computed using cost-distance analysis incorporating topographic travel time via Tobler's hiking function (Tobler 1993; Herzog 2020). Table A5.1 lists each parameter's classification, data source, transformation function type, and calibration values. Parameters excluded from the suitability summation are retained in the model as inputs to resource calculations (arable soils, vegetation, available pasture) or as intermediate derivations, but are not included as independent weighted terms to avoid overrepresenting annual rainfall, from which they are derived.

Parameter	Type	Source / Derivation	Fuzzy Function	Transformation Parameters	Included in Suitability Summation
Mean Annual Rainfall	Permanent Ecological	IMS station data; IDW interpolation	Linear	Min = 0 mm, Max = 100 mm	Yes
Slope	Permanent Ecological	JAXA ALOS PALSAR DEM (12.5 m)	FuzzySmall	Midpoint = 15°, Spread = 3	Yes
Proximity to Arable Soils	Permanent Ecological	Israeli Geological Survey (loess & Hatzeva Fm.); cost-distance (Tobler)	—	—	No (collinear with annual rainfall)
Distance to Permanent Water Sources	Permanent Ecological	Topographic maps & field sources; cost-distance (Tobler)	FuzzySmall	Midpoint = 3 hrs walking, Spread = 1	Yes
Vegetation Type	Permanent Ecological	Danin (1983); Ron et al. (2024)	—	—	No (used in pasture derivation only)
Distance to Tell el-Qudeirat	Permanent Anthropogenic	Site location; cost-distance (Tobler)	FuzzySmall	Midpoint = 4 hrs walking, Spread = 1	Yes
Annual Rainfall	Annual Ecological	IMS historical records (Aug–Jul); IDW interpolation;	Linear	Min = 0 mm, Max = 100 mm	Yes

		min. 20 active stations			
Available Pasture	Annual Ecological	Vegetation type × annual rainfall (multiplicative interaction)	—	—	No (collinear with annual rainfall)
Fertile Arable Soils (Seasonal)	Annual Ecological	Zonal rainfall totals on loess/Hatzeva polygons; cost-distance	—	Rainfall threshold: mean rainfall > 75 mm **and** 25th percentile rainfall > 60 mm	No (collinear with annual rainfall)
Multi-Annual Resource Exhaustion	Dynamic Anthropogenic	Accumulated from simulation: weighted degradation history + focal spatial smoothing	—	Temporal decay factor = 0.5; spatial radius = 5 km	Yes
Reuse Potential	Dynamic Anthropogenic	Accumulated from simulation: encampment and enclosure construction history	—	Recent constructions weighted higher than older ones	Yes

Table A5.1. Environmental parameter specifications: classification, data sources, fuzzy membership transformation functions, and inclusion in the weighted suitability summation. 'Permanent' parameters remain constant across all 75 simulation years; 'Annual' parameters update each year through stochastic rainfall selection; 'Dynamic Anthropogenic' parameters accumulate from simulated household activity.

Notes: FuzzySmall assigns maximum suitability (10) to low input values and minimum suitability (0) to high input values, using a sigmoidal decline; midpoint is the input value at suitability 0.5 and spread controls the steepness of the transition. Linear transformation maps raw values linearly between the specified minimum (suitability 0) and maximum (suitability 10). '—' indicates that no fuzzy transformation is applied because the parameter is either binary, categorical, or derived through multiplicative interaction rather than normalised independently

A5.2 Degradation and Carrying Capacity Equations

Household occupation degrades environmental suitability through two interacting mechanisms that together produce the landscape patchiness driving temporal cycling (Section 5.3.1). The first operates *within* each annual cycle: every time a household or nuclear family camp establishes a location, suitability values in the surrounding cells are reduced immediately, so that households placing later in the same year encounter an already-modified landscape. The second operates *across* years: each year's occupation pattern is added to a weighted cumulative record, so that areas with a longer history of repeated use carry persistent degradation that outlasts any single season.

Equations eq A5.1 and eq A5.2 formalise the within-year spatial degradation. Both apply identically in structure, differing only in the magnitude of the penalty term (D for household territories, P for nuclear family camps) and the spatial radius over which they operate.

$$eq\ 4.1: S'(x,y) = \max\left(S(x,y) - \frac{D}{d(x,y)}, 0.0001\right)$$

Equations A5.1 and A5.2. Distance-based environmental degradation from household and nuclear family occupation. Where S'(x,y) is the updated suitability at cell (x,y), S(x,y) is the original suitability, D is the degradation factor, d(x,y) is Manhattan distance from the occupied cell, and P is the position degradation factor for the centre cell. Households use D = 0.5 and P = 0.9; nuclear family camps use D = 0.3 and P = 0.5. Minimum suitability is 0.0001.

$$eq\ 4.2: S'(center) = \max(S(center) - P, 0.0001)$$

Equation eq A5.3 formalises the temporal decay function that weights each past year's contribution to the cumulative degradation record. The decay factor of 0.5 means that occupation two years ago exerts half the influence of occupation last year, and occupation five years ago exerts less than one-sixteenth of last year's influence, implementing an implicit landscape recovery process without requiring a separate recovery model.

$$eq\ 4.3 + 2: w(t) = 0.5^{T-t}$$

Table A5.2. Notation for degradation equations.

Equation A5.3. Temporal decay weighting for inter-annual degradation. w(t) is the weight applied to occupation in year t, T is the current year, and (T-t) is years since occupation. The decay factor of 0.5 weights recent occupation more heavily, implementing implicit landscape recovery over time.

Symbol	Description
$S(x,y)$	Suitability of cell (x, y)before degradation
$S'(x,y)$	Suitability of cell (x, y)after degradation
D	Degradation intensity parameter for household territory (fixed at 0.5, weight 0.9)
P	Direct centre penalty for nuclear family camp occupation

$d(x, y)$	Distance from occupation centre to cell (x, y)in cell units
$w(t)$	Temporal weight assigned to occupation in year t
T	Current simulation year
t	Year of past occupation being weighted

Cumulative weighted degradation from all past years is subsequently smoothed through spatial convolution over a 5 km radius using inverse-distance weighting, spreading degradation pressure beyond directly occupied cells to represent landscape-scale grazing and resource extraction impacts. The resulting stress surface feeds directly into the annual suitability raster as the "multi-annual resource exhaustion" parameter (Table A5.1), contributing to location selection decisions in subsequent years.

A5.3 Calibration Implementation Details

Calibration proceeded in two phases. Prior to formal optimisation, exploratory manual runs were conducted to develop intuition about model sensitivity and verify basic model behaviour. These initial runs confirmed that the model could produce qualitatively plausible outputs, identified parameters with strong versus weak effects on pattern matching, and resolved technical implementation issues before committing computational resources to systematic optimisation. Exploratory runs revealed, for example, that annual rainfall weights required substantial influence to generate appropriate temporal cycling, while some parameters initially considered were later excluded as redundant given their minimal additional pattern-matching contribution.

Formal optimisation employed Optuna (Akiba et al. 2019) with the Tree-structured Parzen Estimator (TPE) sampler for adaptive parameter space exploration (Watanabe 2023). TPE constructs probabilistic models of promising versus unpromising regions of parameter space based on previous trial outcomes, concentrating subsequent sampling in areas likely to yield improved objective scores. A MedianPruner terminated trials early when intermediate results indicated unlikely improvement over previous trials, reducing computational waste on unpromising parameter combinations. Together, these algorithmic choices balance exploration of diverse parameter combinations against exploitation of promising regions.

Formal optimisation executed 200 trials (143 completing fully and 57 pruned by early stopping). With four workers running trials in parallel, each completing trial required a mean of approximately 8.4 hours of wall-clock time (range: 3.0–19.8 hours), reflecting the ten-replicate structure and 75-year simulation length per replicate. Total wall-clock computation across the optimisation campaign spanned approximately 12–13 days.

Reproducibility. Random seeds were generated deterministically as $seed = (t \times 100) + i$, where $t$is the trial number and $i$is the replicate index, ensuring full reproducibility while maintaining independence between replicates within each trial. The seven integer weights sampled during each trial were normalised to proportional values by dividing each by their sum before application; for example, sampled values [3, 0, 5, 2, 1, 4, 6] normalise to [0.143, 0.000, 0.238, 0.095, 0.048, 0.190, 0.286], ensuring weights represent relative environmental preferences rather than absolute scaling factors.

A5.4 Full Model Parameter Reference

Table A5.4 provides a comprehensive reference for all fixed and calibrated parameters used in the model. Parameters are grouped by functional category. 'Assumed' in the Source column indicates that no direct empirical basis was available for the specific value; these parameters were set through logical reasoning about pastoral systems and model behaviour, and their influence on outputs is assessed through sensitivity analysis (Section 5.6). Parameters listed under 'Calibrated' in the Environmental Suitability Weights group were determined through Optuna optimisation as described in Sections 5.5.2 and A5.3; their values are reported in Section 5.5.4.

Parameter	Value / Range	Units	Description	Source / Justification	Section
Spatial Configuration					
Cell size	250 × 250	m (6.25 ha)	Spatial resolution of grid cells	Balances computational efficiency and environmental heterogeneity; aligns with survey coordinate accuracy (~100–500 m)	5.2.1
Grid dimensions	318 × 280	cells	Total model extent (3,251.54 km²)	Natural topographic boundaries of Negev Highlands; western limit imposed by Israel–	5.2.1

				Egypt border (data constraint)	
Environmental Suitability Weights (calibrated via Optuna — see A5.3)					
Mean annual rainfall weight	0–7 (integer, calibrated)	relative weight	Importance of long-term rainfall in location selection	IMS station data; IDW interpolation	5.2.1.2, 5.5.2
Slope suitability weight	0–7 (integer, calibrated)	relative weight	Importance of terrain slope in location selection	JAXA ALOS PALSAR DEM (12.5 m)	5.2.1.2, 5.5.2
Distance to Tell el-Qudeirat weight	0–7 (integer, calibrated)	relative weight	Importance of proximity to central place	Tobler's hiking function (cost-distance)	5.2.1.2, 5.5.2
Distance to permanent water weight	0–7 (integer, calibrated)	relative weight	Importance of water access in location selection	Tobler's hiking function (cost-distance)	5.2.1.2, 5.5.2
Annual rainfall weight	0–7 (integer, calibrated)	relative weight	Importance of current-year rainfall	IMS observed years; IDW interpolation	5.2.1.2, 5.5.2
Reuse potential weight	0–7 (integer, calibrated)	relative weight	Importance of previously occupied sites	Model-generated from occupation history	5.2.1.2, 5.5.2
Multi-annual stress weight	0–7 (integer, calibrated)	relative weight	Importance of cumulative degradation avoidance	Model-generated (see A5.2)	5.2.1.2, 5.5.2
Degradation Parameters					
Household degradation intensity (D)	0.5	dimensionless	Distance-based degradation intensity for household territories	Assumed	5.3.1
Household centre penalty (P)	0.9	dimensionless	Direct centre-cell degradation intensity for household camp	Assumed	5.3.1
Nuclear family degradation intensity (D)	0.3	dimensionless	Distance-based degradation intensity for nuclear family camps	Assumed	5.3.1
Nuclear family centre penalty (P)	0.5	dimensionless	Direct centre-cell degradation intensity for nuclear family camp	Assumed	5.3.1
Temporal decay factor	0.5	dimensionless	Exponential decay weight for cumulative degradation by year	Assumed (implicit recovery rate)	5.3.1
Spatial smoothing radius	5	km (20 cells)	Diffusion distance for cumulative degradation	Assumed (grazing impact range)	5.3.1
Minimum suitability floor	0.0001	dimensionless	Floor value preventing complete suitability collapse	Assumed	5.3.1
Carrying Capacity					
Medium pasture quality reference	5	0–10 scale	Reference suitability value for carrying capacity calculation	Literature-based standard	5.3.2
Pasture carrying capacity	1.125	goats / cell	Livestock support per 250 m cell at medium quality pasture	Ethnographic (Rosen & Finkelstein 1992)	5.3.2
Carrying capacity evaluation radius	25	cells (~6.25 km)	Spatial zone for pastoral productivity assessment	Assumed	5.3.2, 5.3.2
Household Economic Parameters					

Initial flock size	~105 per nuclear family (~1,050 per household)	goats	Starting livestock holdings	Modified from Rosen & Finkelstein (1992)	5.2.2, 5.3.2
Livestock per capita subsistence requirement	18	goats / person	Minimum herd per person for subsistence (diversified economy)	Modified from Rosen & Finkelstein (1992)	5.2.2, 5.3.2
Per capita consumption	1.2	surplus units / person / year	Non-subsistence household expenses	Assumed	5.3.2
Agricultural offset	Up to 40%	% reduction	Reduction in livestock subsistence requirement where agriculture possible	Rosen & Finkelstein (1992); Günther et al. (2021)	5.3.2
Stress maintenance cost	15% × stress ratio	% of herd	Penalty when herd exceeds local carrying capacity	Assumed	5.3.2
Annual surplus decay	30%	% per year	Baseline resource spoilage and gradual consumption	Assumed	5.3.2
Large surplus decay (progressive)	5–25%	% per year	Additional decay above 100 surplus units (storage limits, redistribution pressure)	Assumed	5.3.2
Maximum herd growth rate	Up to 20%	% per year	Annual herd growth under favourable conditions; diminishing returns approaching 1,000 animals	Literature-based	5.3.2
Herd decline rate	10–30%	% per year	Decline when carrying capacity insufficient; scaled by resource deficit	Literature-based	5.3.2
Maximum herd size	1,000	animals	Upper cap for herd growth (diminishing returns threshold)	Assumed	5.3.2
Crisis Response Parameters					
Crisis surplus threshold	10	surplus units	Surplus level triggering emergency livestock conversion	Assumed	5.3.4
Emergency livestock conversion rate	1 livestock → 0.3 surplus	ratio	Unfavourable distressed-sale conversion rate	Assumed	5.3.4
Minimum reserve herd (crisis)	18 animals / person, capped at 100	animals	Minimum herd protected during emergency conversion	Assumed	5.3.4
Livestock acquisition threshold	Surplus >80 and herd <600	units / animals	Conditions triggering opportunistic herd rebuilding; invests up to 20% of surplus, max 40 units	Assumed	5.3.4
Livestock acquisition rate	0.3 surplus → 1 livestock	ratio	Premium cost of purchasing animals	Assumed	5.3.4

Household extinction: surplus	<0	surplus units	Household failure threshold (surplus)	Assumed	5.3.4
Household extinction: livestock	<15	animals	Household failure threshold (herd)	Assumed	5.3.4
Spatial and Territorial Parameters					
Base territory radius	20	cells (~5 km)	Standard territory size under favourable conditions (quality >5.0)	Assumed (pastoral ranging)	5.3.1
Expanded territory radius (poor conditions)	30	cells (~7.5 km)	Territory when environmental quality <3.5	Assumed (increased ranging needs)	5.3.1
Moderate territory radius	25	cells (~6.25 km)	Territory when environmental quality 3.5–5.0	Assumed	5.3.1
Territorial overlap threshold	25%	% of cells	Maximum acceptable territory overlap before location rejected	Assumed (cooperative coexistence)	5.3.1, 5.3.1
Nuclear family camp radius	11	cells (~2.75 km)	Impact zone for nuclear family sub-camps	Assumed	5.3.1
Enclosure Construction Parameters					
Base prosperity threshold (enclosure eligibility)	0.6	dimensionless	Minimum prosperity index to qualify for enclosure activity	Assumed	5.3.3
High prosperity threshold (new construction)	0.8	dimensionless	Minimum prosperity index to build a new enclosure	Assumed	5.3.3
Minimum manpower — reuse	≥15	persons	Required workforce to reuse existing enclosure	Assumed	5.3.3
Minimum manpower — new construction	≥30	persons	Required workforce to build new enclosure	Assumed	5.3.3
Enclosure reuse cost (manpower)	10	persons	Labour investment to reuse existing structure	Assumed	5.3.3
Enclosure reuse cost (surplus)	6	surplus units	Resource investment to reuse existing structure	Assumed	5.3.3
New enclosure cost (manpower)	20	persons	Labour investment to construct new enclosure	Assumed	5.3.3
New enclosure cost (surplus)	25	surplus units	Resource investment to construct new enclosure	Assumed	5.3.3
Manpower recovery after construction	90%	% returned	Proportion of manpower cost recovered after construction completes	Assumed	5.3.3
Enclosure reuse window — own constructions	35	years	Maximum age of household's own enclosures eligible for reuse	Assumed	5.3.3
Enclosure reuse window — others' enclosures	25	years	Maximum age of other households' enclosures eligible for reuse	Assumed	5.3.3

Others' enclosure reuse weighting	80%	% of own weighting	Priority discount for reusing another household's enclosure	Assumed	5.3.3
Enclosure selection stochastic variation	±20%	%	Random variation in enclosure selection probability	Assumed	5.3.3
Memory and Territorial Attachment Parameters					
Territorial centre weighting — previous	90%	% contribution	Weight of previous centre in updating home-range focus	Assumed	5.3.1
Territorial centre weighting — current	10%	% contribution	Weight of current year's location in updating home-range focus	Assumed	5.3.1
Territorial centre Gaussian peak bonus	+5.0	suitability units	Maximum suitability bonus at home-range centre	Assumed	5.3.1
Territorial centre Gaussian sigma	50	cells	Spatial spread of home-range centre attraction	Assumed	5.3.1
Territory boundary bonus	2.5 / (distance + 1)	suitability units	Distance-based bonus for cells near previous year's territory (up to 20 cells beyond edge)	Assumed	5.3.1
Memory quality threshold	5	0–10 scale	Quality value distinguishing positive from negative memory effects	Assumed	5.3.1
Current-year location penalty	−0.5	suitability units	Fixed penalty preventing immediate re-occupation of current year's location	Assumed	5.3.1
Memory guaranteed retention	<15	years	Memories younger than 15 years are always kept	Assumed (intergenerational knowledge)	5.3.1
Memory median lifespan	~19–20	years	Older memories dropped stochastically (annual retention probability ≈ 15 ÷ age); few persist beyond ~two decades	Assumed	5.3.1
Temporal Parameters					
Simulation duration	75	years	Total modelled timespan (c. 980–830 BCE)	Iron Age IIA system duration (Ben-Yosef et al. 2021; Ben-Yosef 2023)	5.2.3
Time step	1	year	Temporal resolution; each step represents one summer occupation season	Annual seasonal cycle	5.2.3
Initial population	9	households	Starting number of household agents	Conservative carrying capacity estimate (Rosen & Finkelstein 1992); see Section 5.2.4	5.2.4

Table A5.4. Full model parameter reference. Parameters are grouped by functional category. 'Assumed' indicates values set by logical reasoning or model design in the absence of direct empirical data. Calibrated suitability weights are reported in Section 5.5.4

A5.5 Household Agent State Variables

Table A5.5 lists all state variables maintained by individual household agents throughout the simulation. Dynamic variables update during each annual cycle; static variables are set at initialization and remain fixed. Variable names correspond to implementation in the Mesa-based Python model (ter Hoeven et al. 2025).

Variable	Type	Units	Range	Dynamic / Static	Description	Section
flock_head	Integer	animals (goats)	≥15	Dynamic	Total livestock holdings	5.2.2, 5.3.2
surplus	Float	abstract resource units	≥0	Dynamic	Accumulated economic resources available for investment and consumption	5.2.2, 5.3.2
manpower	Integer	persons	≥1	Dynamic	Available workforce for herding, movement, and construction	5.2.2
territory	List of coordinates	cell positions	Variable extent	Dynamic (annual reset)	Claimed spatial area; resets each year following repositioning	5.2.2, 5.3.1
memory	List of tuples	[position, quality, year]	Variable length	Dynamic	Personal encampment history; subject to stochastic annual pruning	5.2.2, 5.3.1
enc_memory	List of tuples	[position, year]	Variable length	Dynamic	Collective enclosure knowledge; includes own constructions and encountered enclosures	5.2.2, 5.3.3
in_severe_crisis	Boolean	True / False	Binary	Dynamic	Crisis status flag; triggers emergency livestock conversion when True	5.2.2, 5.3.4
own_suitability_raster	NumPy array	0–10+ scale	318 × 280 cells	Dynamic	Personalised landscape perception: global suitability modified by territorial attachment and	5.2.2, 5.3.1

					memory adjustments	
pos	Tuple (x, y)	grid coordinates	Within model bounds	Dynamic (annual reset)	Current camp location; reset to None at year end and re-selected at year start	5.2.2

Table A5.5. Household agent state variables. All variables are dynamic unless noted. Variable names reflect Python implementation.

A5.6 Environment and Model-Level State Variables

Table A5.6 lists all environment-level and model-level state variables maintained by the simulation. Static rasters are loaded once at initialization from pre-processed GIS datasets; dynamic rasters update each annual cycle following household decision-making. All rasters share the 318 × 280 cell grid defined in Section 5.2.1.

Variable	Type	Units	Range	Dynamic / Static	Description	Section
year	Integer	years	0–74	Dynamic	Current simulation year; drives annual cycle execution and stochastic rainfall selection	5.2.3
suitability_raster	2D NumPy array	0–10+ scale	318 × 280	Dynamic (annual update)	Overall environmental suitability combining all weighted parameters; drives household location selection	5.2.1.2
return_raster	2D NumPy array	0+ scale	318 × 280	Dynamic (annual update)	Reuse potential surface accumulating past encampment and enclosure history; weighted toward recent constructions	5.2.1.2
stress_ras	2D NumPy array	0–10 scale	318 × 280	Dynamic (annual update)	Cumulative human degradation pressure; weighted temporal accumulation spatially smoothed over 5 km radius (see A5.2)	5.2.1.2, 5.3.1
veg_ras	2D NumPy array	0–10 scale	318 × 280	Dynamic (annual update)	Annual vegetation/pasture suitability; multiplicative interaction of vegetation type	5.2.1.2

					and selected rainfall year	
ag_ras	2D NumPy array	0–10 scale	318 × 280	Dynamic (annual update)	Annual agricultural potential; cost-distance to seasonally cultivable soils given selected rainfall year	5.2.1.2
target_raster	2D NumPy array	0+ (cumulative counts)	318 × 280	Dynamic (annual accumulation)	Spatial activity tracker accumulating household occupation signals across all 75 years; forms basis for simulated site classification	5.2.1.2
place_raster	2D NumPy array	0 / 1 (binary)	318 × 280	Static	Valid placement mask; restricts agent activity to Israeli portion of Negev Highlands where environmental data coverage is complete	5.2.1, 5.2.1.2
enclosures	List	[position, year, agent_id]	Variable length	Dynamic	Registry of all constructed enclosures; queried during enclosure search and reuse decisions	5.2.1.2, 5.3.3
territories	List of lists	nested cell coordinates	Variable length	Dynamic (annual reset)	Registry of all active household territories for the current year; used in overlap evaluation	5.2.1.2, 5.3.1
mean_annual_rainfall	2D NumPy array	mm	318 × 280	Static	Long-term average precipitation derived from IMS station data via IDW interpolation	5.2.1.2
slope	2D NumPy array	degrees	318 × 280	Static	Terrain slope derived from JAXA ALOS PALSAR DEM (12.5 m)	5.2.1.2
kb_distance	2D NumPy array	hours walking	318 × 280	Static	Cost-distance to Tell el-Qudeirat via Tobler's hiking function	5.2.1.2
pw_distance	2D NumPy array	hours walking	318 × 280	Static	Cost-distance to nearest permanent water source via Tobler's hiking function	5.2.1.2
annual_rainfall	2D NumPy array	mm	318 × 280	Dynamic (annual update)	Stochastically selected historical rainfall year; updated each simulation year from IMS records	5.2.1.2

arable_soil_distance	2D NumPy array	hours walking	318 × 280	Dynamic (annual update)	Cost-distance to seasonally fertile soils; recalculated each year based on selected rainfall threshold	5.2.1.2

Table A5.6. Environment and model-level state variables. Static variables are set at initialization; dynamic variables update each simulation year. All 2D NumPy arrays share the 318 × 280 cell spatial extent.

A5.7 Sensitivity Analysis — Full Results

This appendix reports the complete methodological specifications, numeric results, and spatial figures for all four sensitivity tests described in Section 5.6. Tests are presented in the order discussed in the main text.

A5.7.1 Null Weight Baseline

To establish a performance baseline against which the calibrated solution could be evaluated, 30 model iterations were executed using a null weight configuration in which all seven environmental suitability parameters received equal weights (each $1/7 \approx 0.143$). This configuration represents the absence of any preferential environmental structure in household location selection: no parameter is considered more or less important than any other. The null design deliberately excludes the calibrated preference structure to test whether the optimisation process identified a genuinely informative parameter combination or merely converged on a solution achievable by any weighting scheme. Each iteration used a distinct random seed following the same deterministic seed generation procedure as calibration (seed = trial × 100 + iteration). Performance was measured using the same composite objective function as calibration (spatial error + ratio error / 2).

Statistic	Value
Mean error	0.3182
95% CI lower	0.3088
95% CI upper	0.3277
Minimum	~0.28
Maximum	~0.39
Best calibrated	0.2013
Improvement	36.7%

Table A5.7. Results of the null weight baseline test

The best calibrated trial score of 0.2013 falls entirely outside the null distribution (Figure A5.1). The gap of 0.117 error units between the calibrated solution and the null mean confirms that the optimised parameter structure captures meaningful environmental signal that an uninformed equal-weight model cannot replicate.

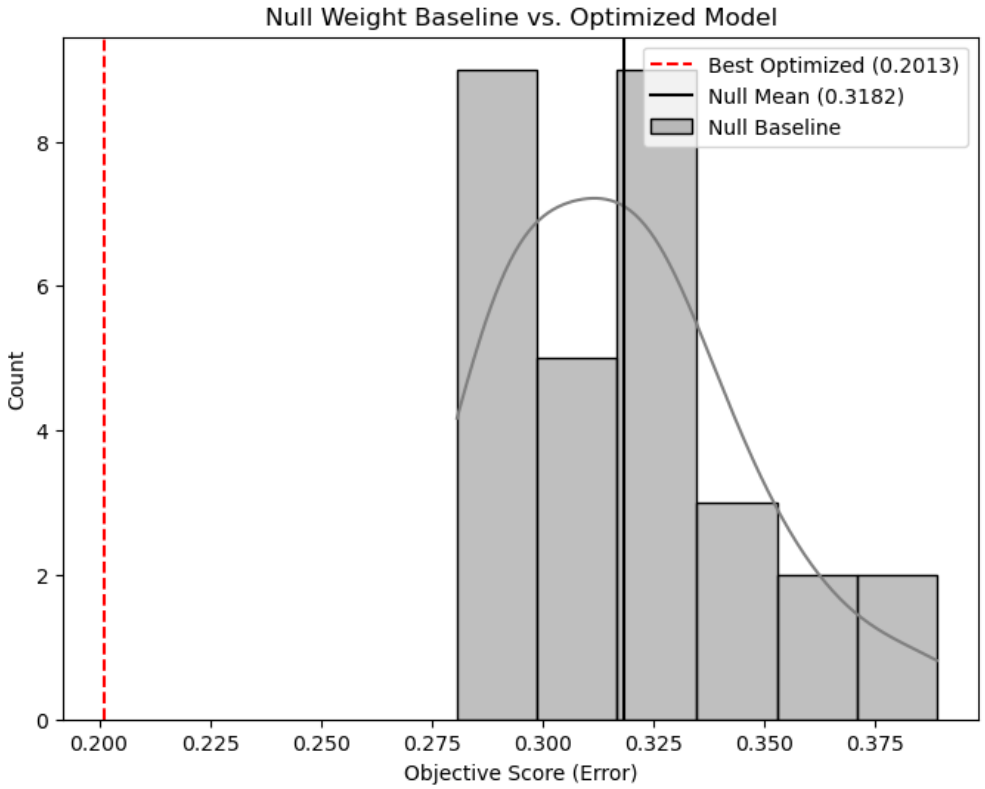


Figure A5.1. Histogram of null baseline composite scores (n=30) with best calibrated score (0.2013) marked as vertical dashed line. Null mean (0.3182) marked separately.

A5.7.2 Temporal Horizon

To test whether the 75-year simulation duration represents an appropriate and stable configuration, a single extended run was executed for 150 years using the calibrated weights from the best trial. Model state was evaluated at ten-year intervals by converting the accumulated target raster to a GeoDataFrame and computing the composite objective score against the archaeological calibration dataset. Three metrics were recorded at each checkpoint: total composite score, spatial error (1 − IoU of standard deviational ellipses), and ratio error (absolute difference in enclosed compound proportions). A single run rather than multiple iterations was used for this test given its exploratory diagnostic purpose and the substantial computational cost of 150-year simulations.

Spatial outputs were additionally inspected visually at years 100 and 150 to assess whether metric trends corresponded to archaeologically plausible spatial distributions, following the visual plausibility criterion discussed in Section 5.6.

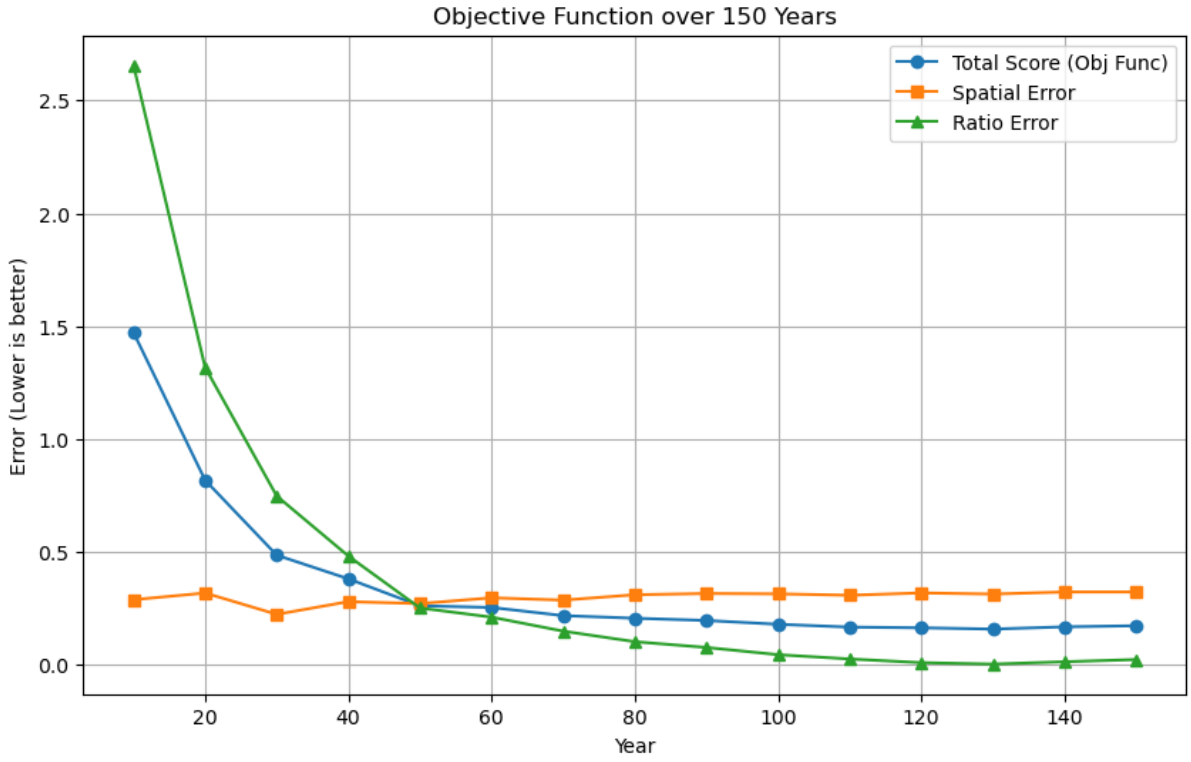


Figure A5.2. Line plot showing total composite score, spatial error, and ratio error across all 15 checkpoints from year 10 to year 150.

Composite error declined steeply from year 10 to year 75, driven primarily by ratio error convergence as enclosure construction events accumulated. Spatial error remained

relatively stable throughout (0.27–0.32). Beyond year 75, apparent metric improvement is driven entirely by continued ratio error reduction toward near-zero saturation; spatial error begins a gradual upward drift from year 100 onward. Visual inspection of spatial outputs at years 100 and 150 reveals near-complete coverage of the valid placement area by simulated activity, consistent with progressive landscape saturation rather than sharpening spatial correspondence (Figures A5.2–A5.3).

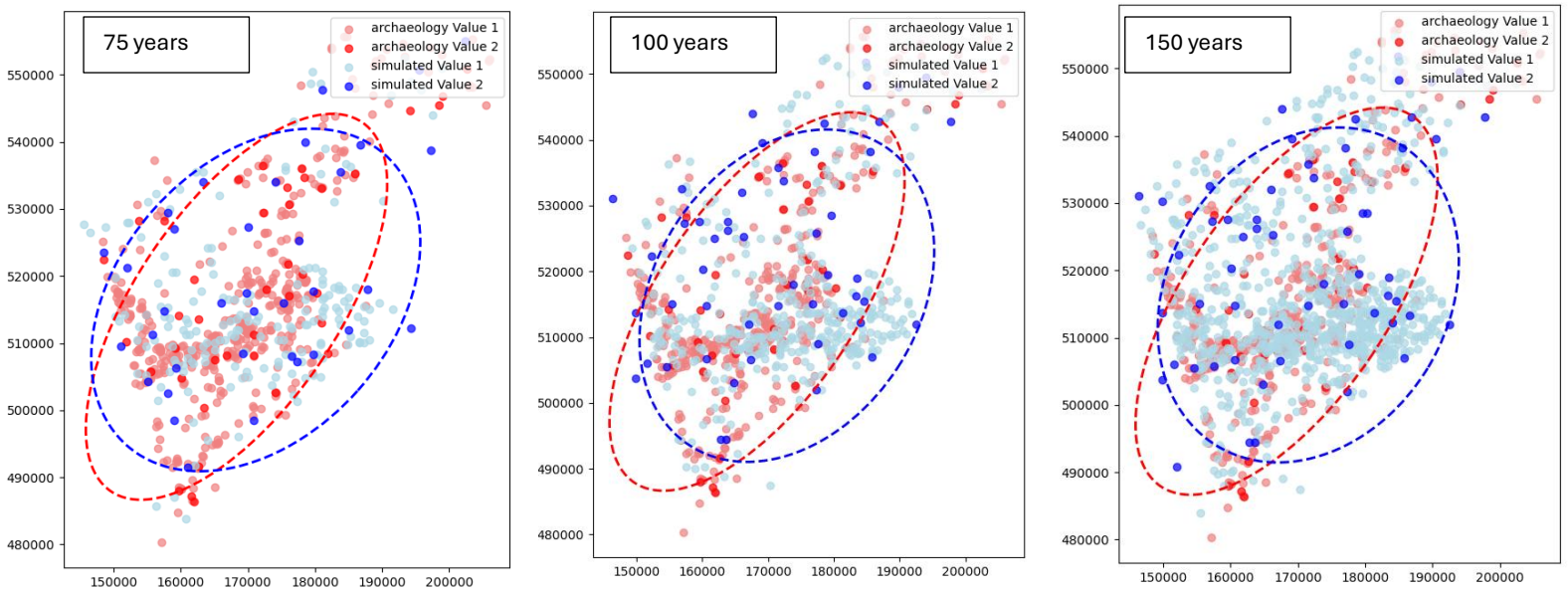


Figure A5.3. Spatial ellipse comparison at years 75, 100, 150 — archaeological distribution (red) versus simulated distribution (blue)

A5.7.3 Population Size

To assess sensitivity to the number of household agents, three alternative population sizes were tested: 5, 15, and 20 agents, compared against the calibrated baseline of 9 agents. Each configuration was run for five iterations using calibrated weights and the standard 75-year duration. Random seeds were assigned using the same deterministic procedure as calibration. Performance was evaluated using the composite objective function. Spatial outputs from each configuration were additionally inspected visually to assess distributional plausibility, given the objective function limitation identified in Section 5.6.

At 5 agents, performance degrades substantially: the sparse household population generates too few occupation events to match the regional archaeological distribution and too few enclosure constructions to approach the 1:13.3 target ratio. At 15 and 20 agents, metric scores improve beyond the nine-agent baseline, but visual inspection reveals this to be artefactual: simulated activity covers practically the entire valid landscape uniformly, with no meaningful spatial clustering corresponding to the observed archaeological concentration (Figures A5.4–A5.5). The nine-agent configuration produces the most spatially coherent output across the tested range.

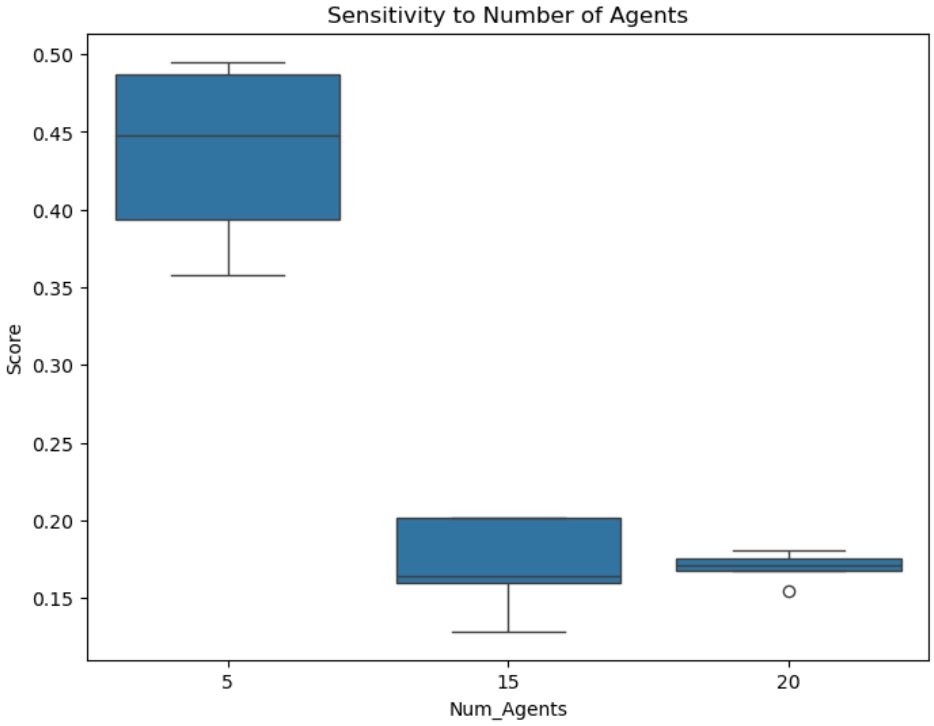


Figure A5.4. Boxplot of composite scores for 5, 15, and 20 agents.

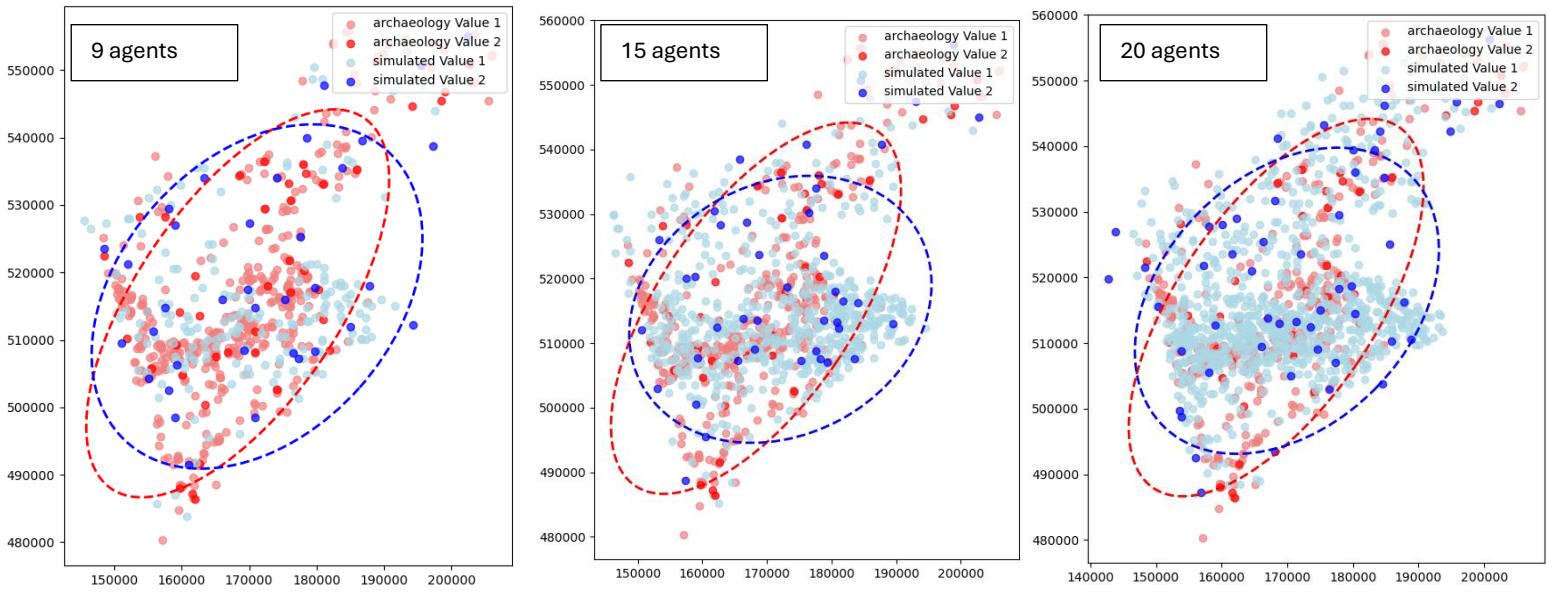


Figure A5.5. Spatial ellipse comparison at 9, 15, 20 agents — archaeological distribution (red) versus simulated distribution (blue)

A5.7.4 Territory Flexibility

To test the contribution of annual landscape-scale repositioning to model performance, two territorial modes were compared. In the default flexible mode, households select new camp locations across the full valid landscape at the start of each year based on updated environmental suitability, accumulated memory, and territorial competition, as described in Section 5.3.1. In the fixed-territory mode, households were assigned territories at initialisation and did not reposition across the landscape in subsequent years; however, within-territory cell selection remained fully operative, meaning households continued to respond to local suitability gradients and degradation within their assigned area. Both modes used identical calibrated weights and the standard 75-year duration. Five iterations were run for each mode using distinct random seeds.

The flexible configuration produced lower mean error and substantially tighter variance. The fixed mode's higher variance reflects dependence on stochastic initial placement: without annual landscape-scale repositioning, outcomes are determined largely by where households happen to be placed at initialisation, producing both near-competitive and substantially degraded results across runs. The considerable distributional overlap between modes reflects the partial behavioural similarity noted in Section 5.6: the fixed

mode retains full intra-territorial flexibility, and households assigned to environmentally suitable initial locations can generate plausible patterns without inter-annual movement.

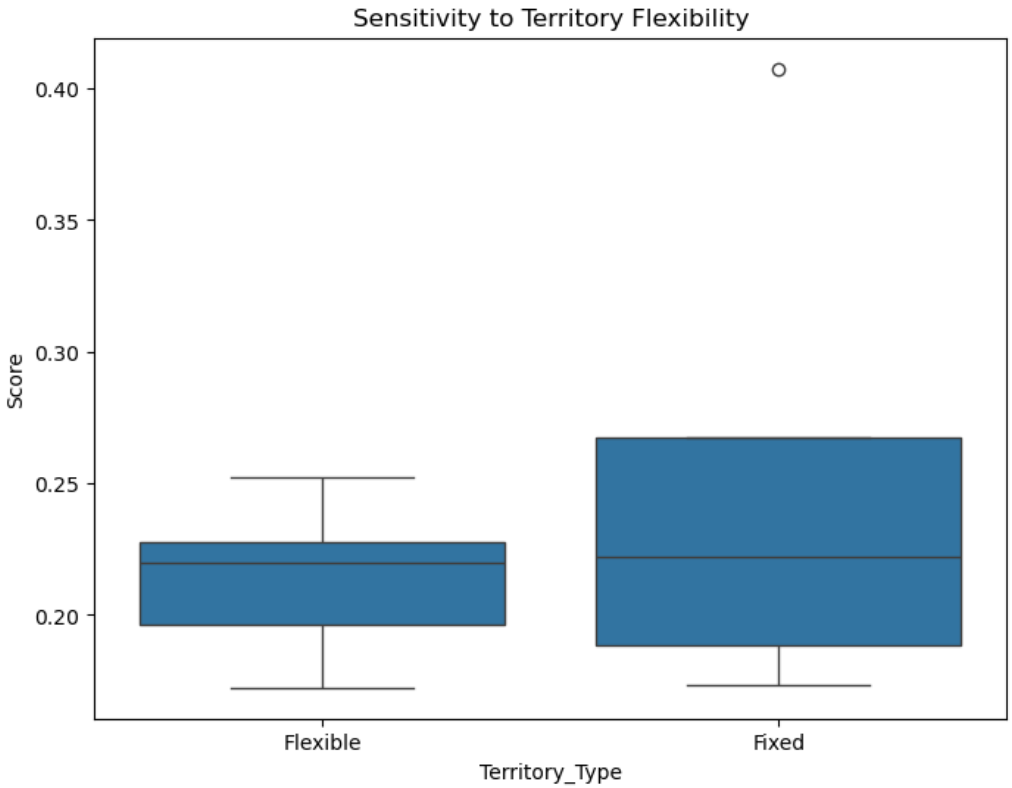


Figure A5.6. Boxplot comparing composite scores for flexible and fixed territory modes across five iterations each.